

Multi-seed M-finescan: vanilla vs orthogonal AL-RNN

120 training runs total: M in {10, 11, 14, 15} x P in {2..6}
x {vanilla, ortho} x 3 seeds, 1000 epochs each, alpha=1.0,
shared frozen B per M, fixed Q_dataset rotation (seed=42).

TL;DR

- * On D_H_state (state-space Hellinger) ortho is statistically significantly better: pooled mean rel gap = -5.83%, bootstrap 95% CI = [-9.66%, -1.78%]; ortho wins on the mean in 16/20 cells; sign-test p = 0.006, Wilcoxon p = 0.006.
- * On PSE (power-spectrum Hellinger) ortho leads directionally but not significantly: pooled mean rel gap = -4.30%, bootstrap 95% CI = [-15.2%, +8.8%]; the CI straddles zero due to two ortho optimisation-failure outliers.
- * On training RMSE (one-step MSE^{0.5}) ortho wins on the mean in 18/20 cells (small but tight).
- * Largest single-cell ortho wins on D_H_state:

M=15	P=4	-20.9%	d_z = -2.55	95% t-CI excludes zero			
M=10	P=5	-12.7%	d_z = -3.05	95% t-CI excludes zero			
M=11	P=4	-14.6%	M=14	P=5	-16.1%, M=15	P=3	-16.9%
- * Parameter-efficiency: ortho-at-M=11 reaches vanilla-at-M=14 quality on PSE (mean) at every P - 3 fewer hidden units for the same Hellinger-attractor quality.

Model architectures

Both models share A (diagonal skip), W (MxM coupling), h (bias) and a frozen input embedding B (3 x M). Only the last P hidden units pass through a ReLU; the first M-P remain linear.

Vanilla AL-RNN

```
z_{t+1} = A * z_t + phi_P(z_t) W^T + h
phi_P(z)[:M-P] = z[:M-P]
phi_P(z)[M-P:] = ReLU(z[M-P:])
```

Orthogonal AL-RNN

```
z_{t+1} = A * z_t + phi_P(Q z_t + b_Q) W^T + h
Q in O(M) enforced by
torch.nn.utils.parametrizations.orthogonal
```

Trainable parameters (frozen B)

	M=10	M=11	M=14	M=15
vanilla	120	143	224	255
ortho	175	209	329	375

Teacher forcing

```
hard reset z[:N] = x at t=0; alpha = 1.0 injection every
n_interleave = 16 steps (z[:N] := (1-alpha)z[:N] + alpha x).
```

Loss

MSE on the first N=3 hidden coords over T=128 steps.

Experimental protocol

Data : Lorenz63 train/test (first 500 steps dropped).
Fixed 3x3 random orthogonal Q_dataset (72 deg, det=+1) rotates the observations in R^3 .
seed_data = 42 (Q_dataset), seed_B = 42 (shared B) are fixed across all runs.

Per-seed : seed_run in {0, 1, 2} controls model init (A, W, h, Q's parametrisation init) AND Python's random.randint used by the dataset's batch sampler.

Optimiser : Adam, lr 1e-3 -> 1e-5 exponentially decayed.

Epochs : 1000 per run; 20 batches/epoch; batch 64.

Seq length : 128; teacher-forcing period n_interleave = 16.

Free-run : after training, 5000-step rollout from X_test_trans[0] with no teacher forcing.

Metrics computed per cell

rmse_train_final - sqrt(final training MSE), one-step
rmse_h100 - free-run RMSE on the first 100 steps
Dstsp - KL($p_{\text{true}} || p_{\text{gen}}$) on a 30^3 binned 3-D density
PSE - Hellinger of smoothed power spectra (per coord, mean)
D_H_state - Hellinger on the same state-space density

Statistical analysis

Per cell (n=3 paired seeds): paired diff o-v, paired SD, Cohen's d_z, paired-t 95% CI, Wilcoxon signed-rank p, win rate over seeds.

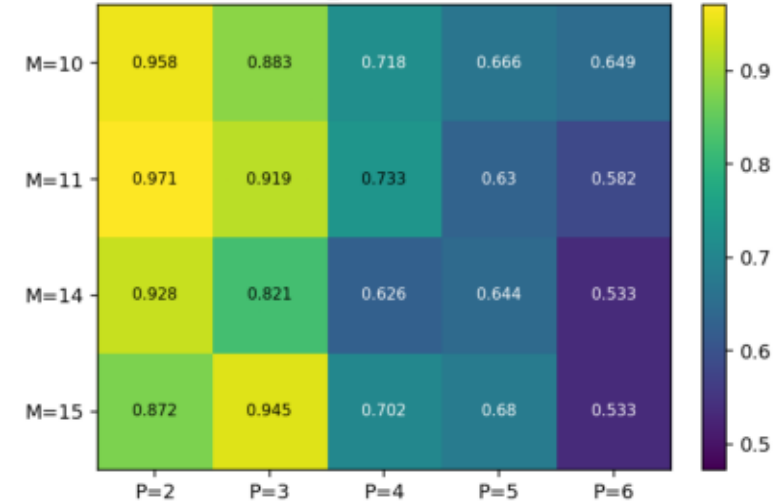
Pooled (20 cells): bootstrap 95% CI on the cell-level mean relative gap, sign test, Wilcoxon on cell-level gaps.

Stratified by M-band (low={10,11} vs high={14,15}) and by P.

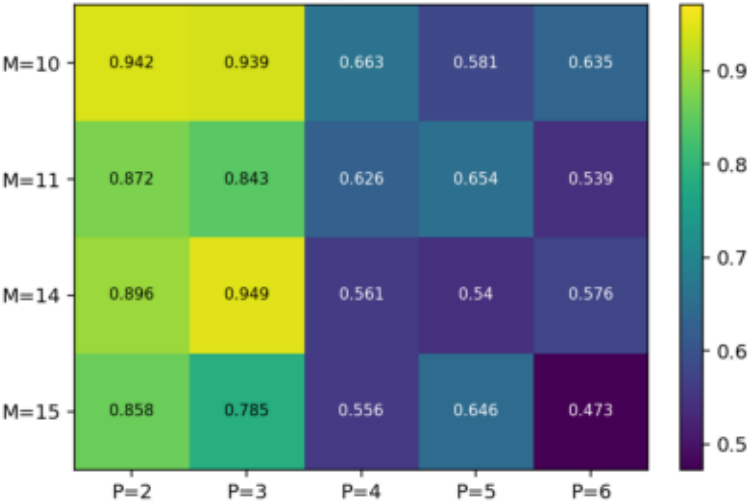
Mean over 3 seeds: D_H state-space Hellinger

D_H state-space (Hellinger) (mean over 3 seeds, lower is better)

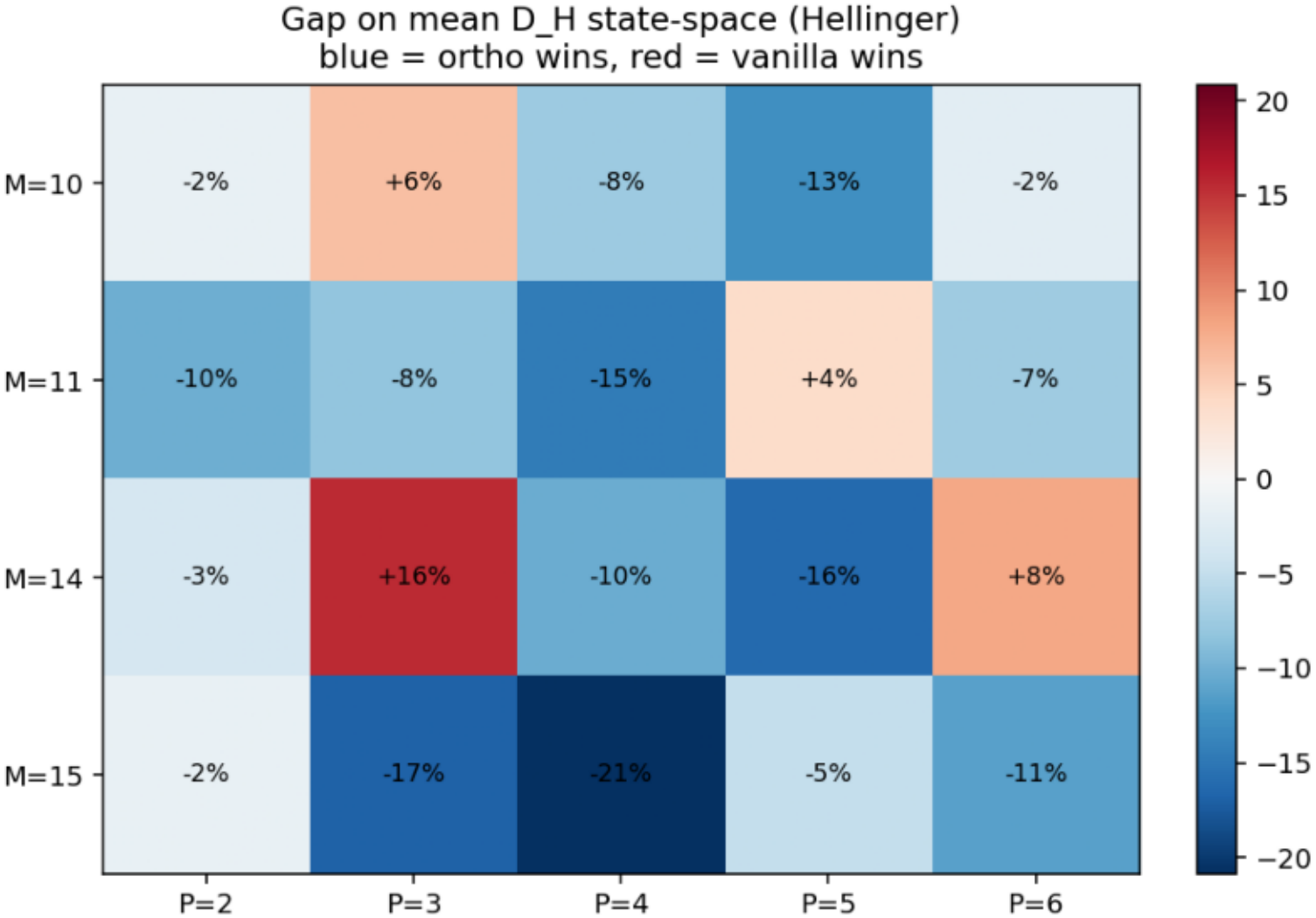
Vanilla mean: D_H state-space (Hellinger)



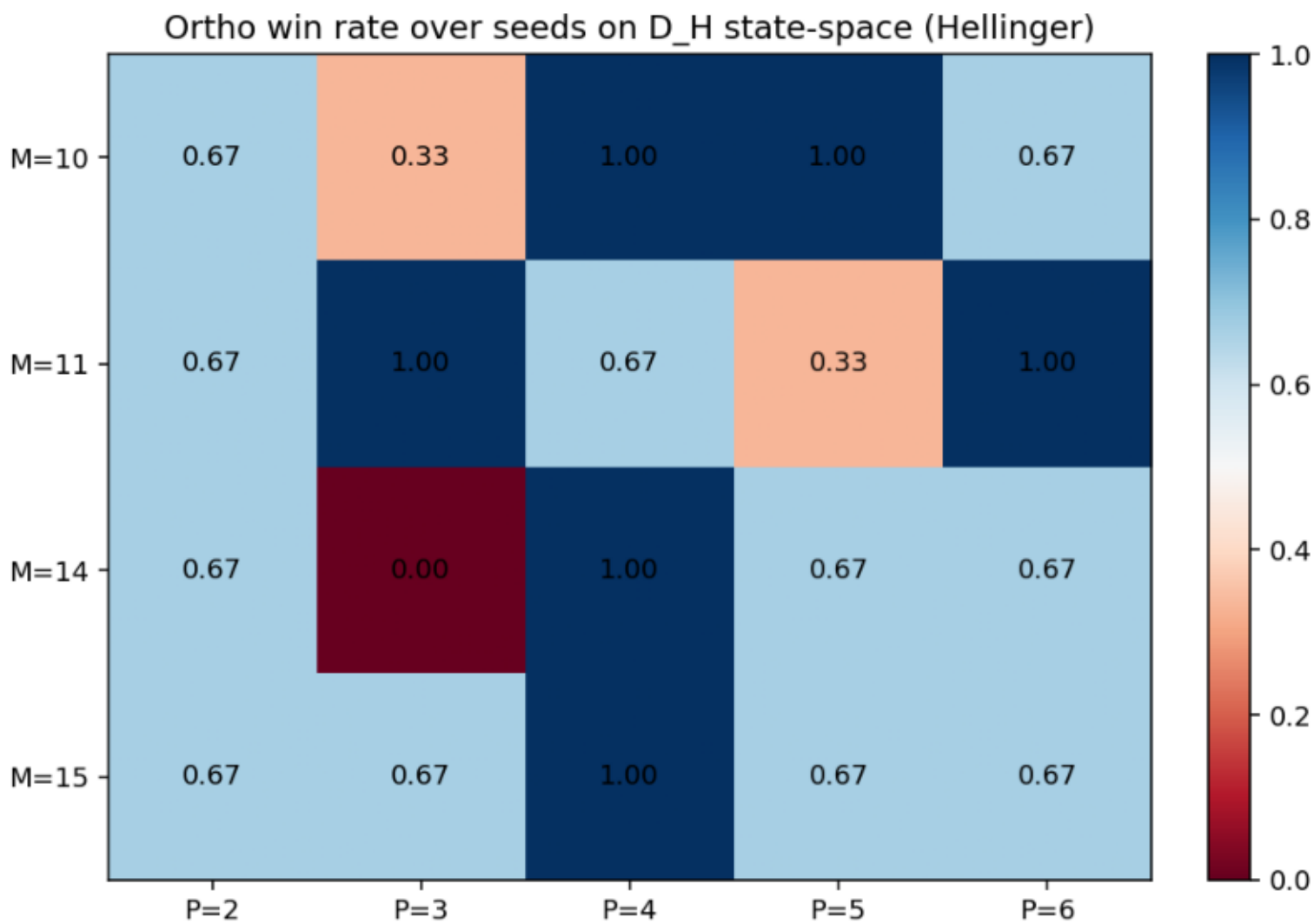
Orthogonal mean: D_H state-space (Hellinger)



Gap (ortho-vanilla)/vanilla on the mean: D_H state-space Hellinger

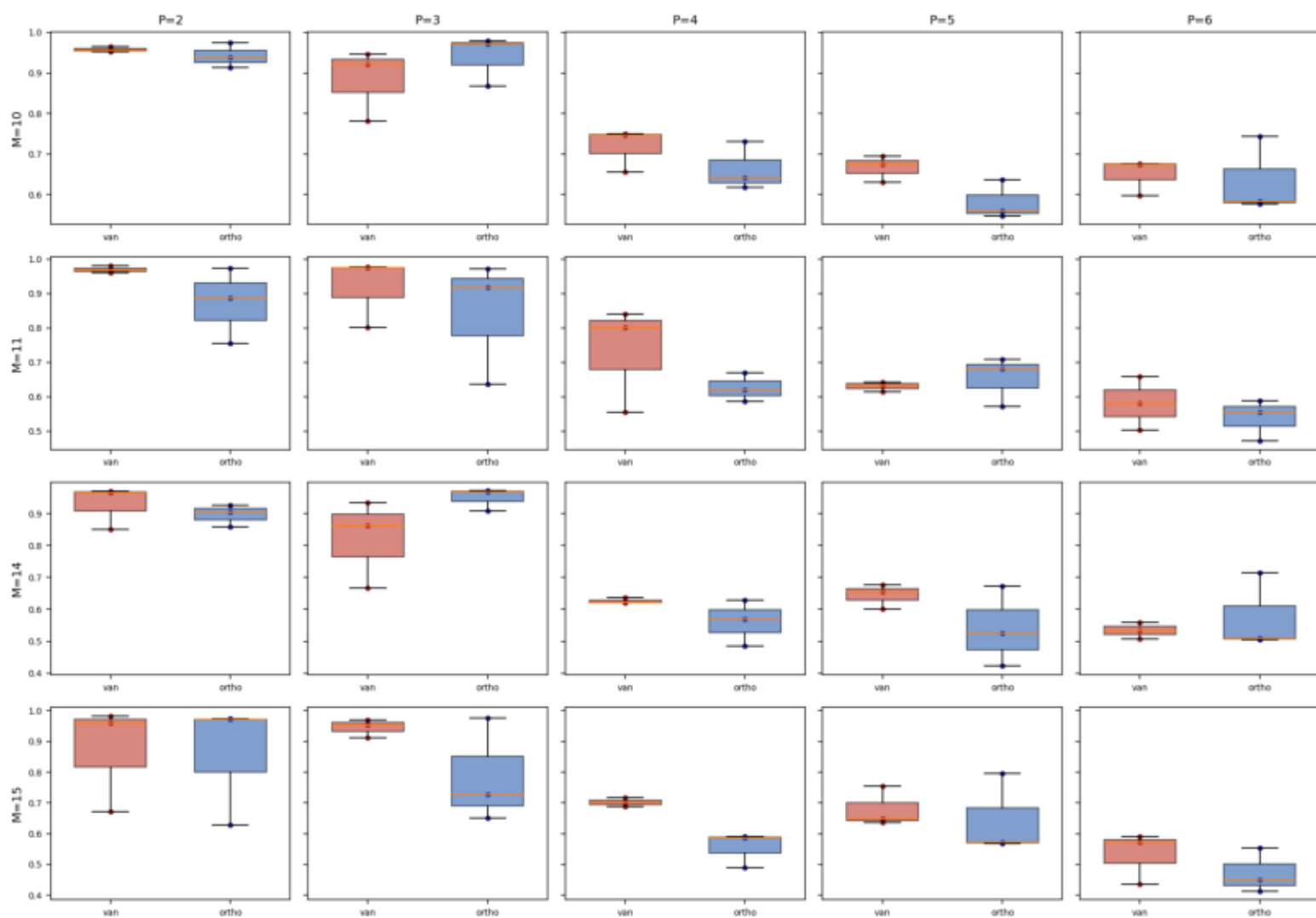


Ortho win rate over 3 seeds: D_H state-space Hellinger



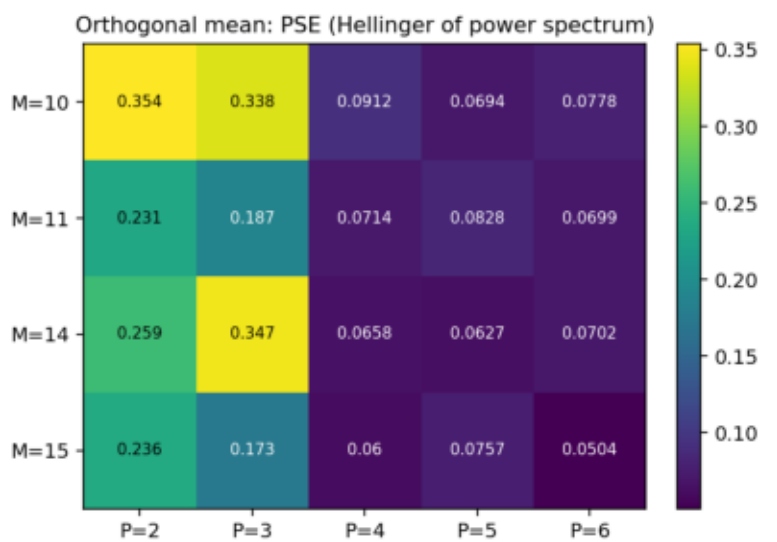
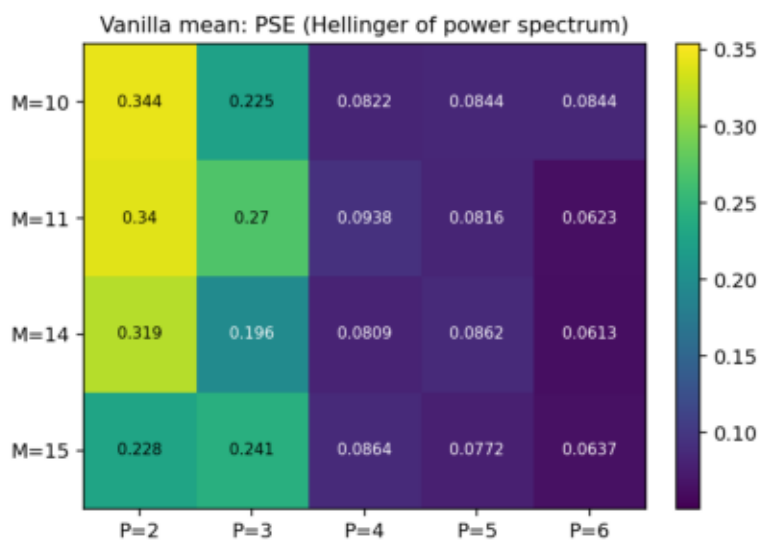
Per-cell box plots over 3 seeds: D_H state-space Hellinger

D_H state-space (Hellinger) (per-seed values; box = IQR, line = median)

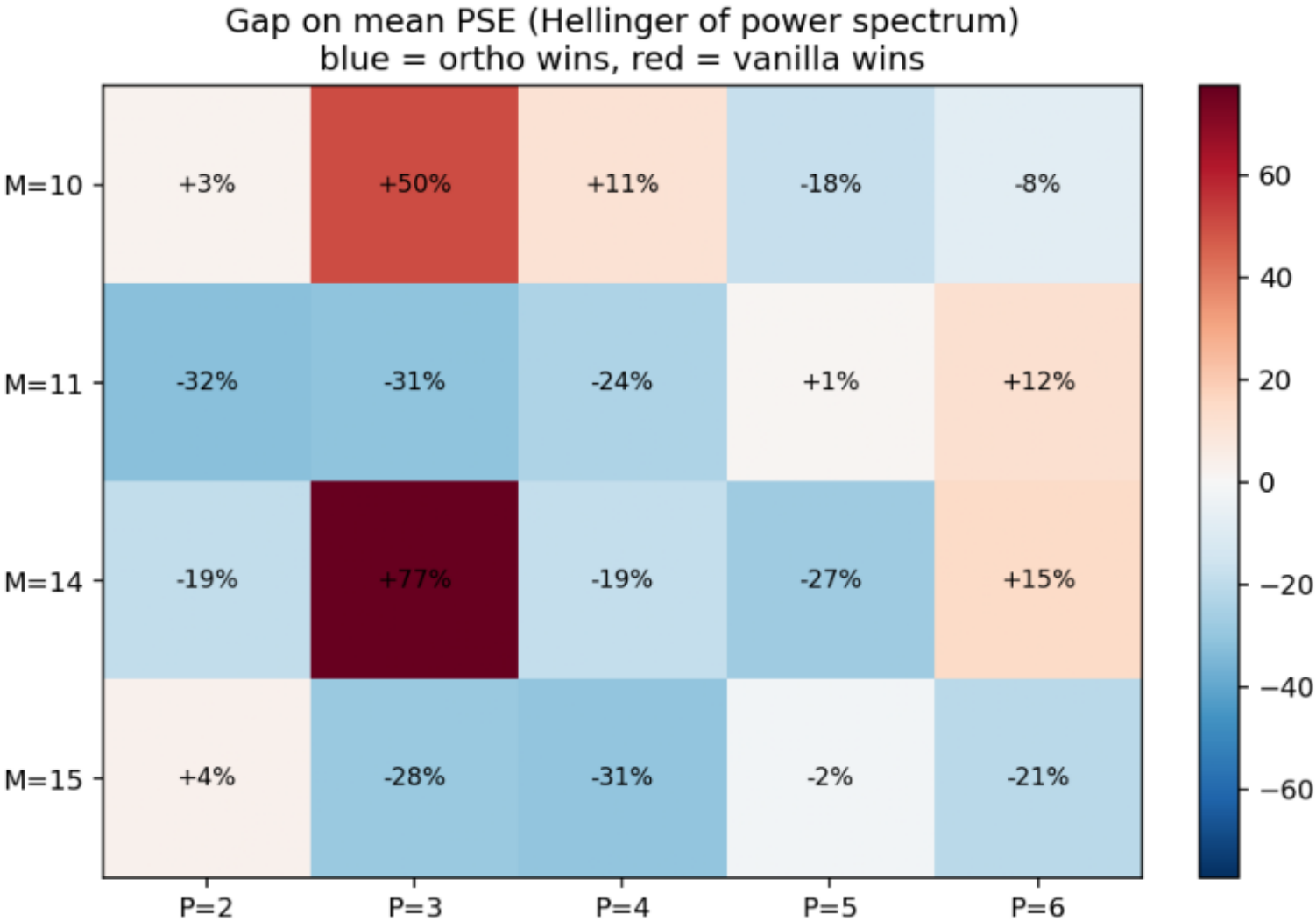


Mean over 3 seeds: PSE (Hellinger of power spectrum)

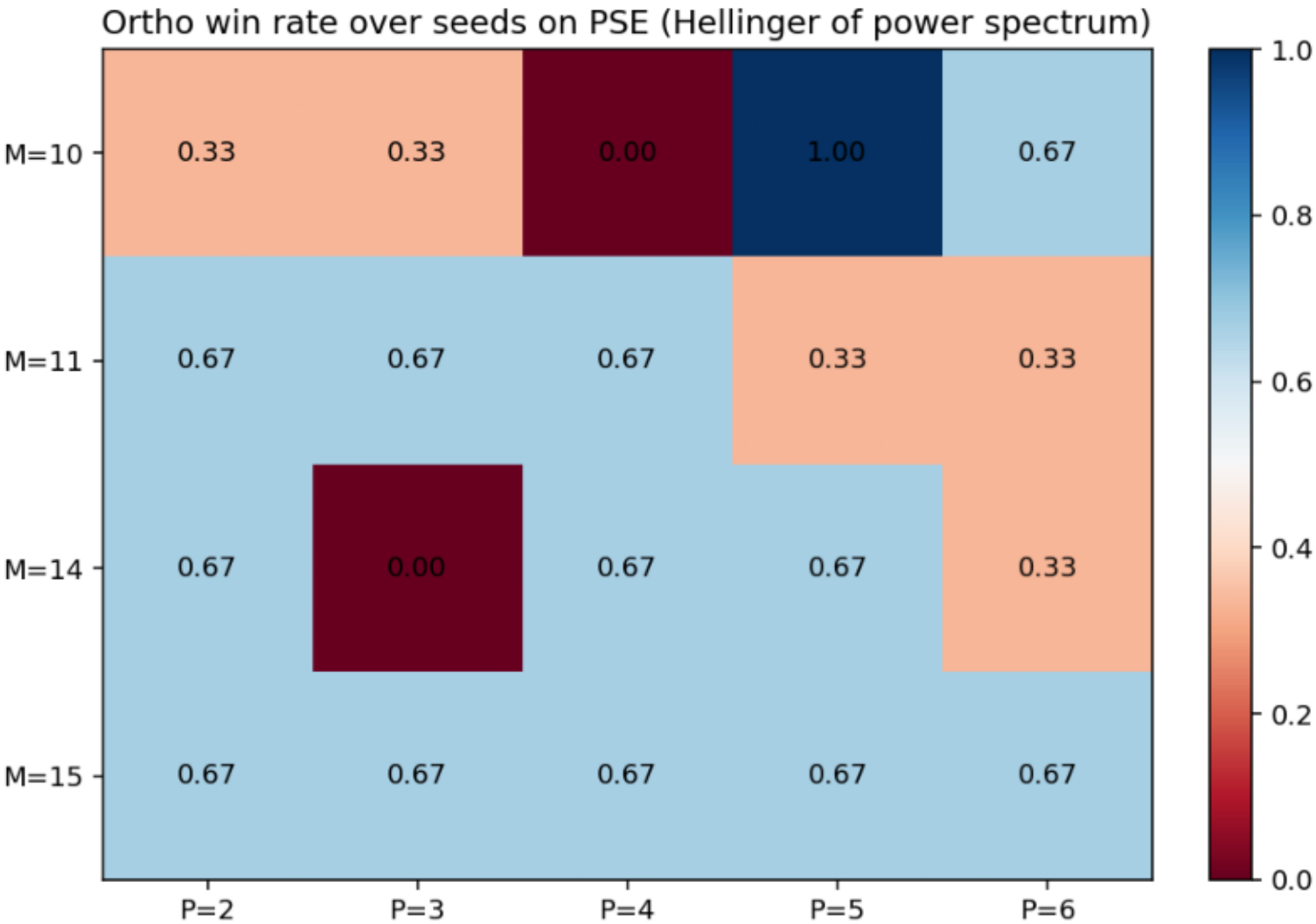
PSE (Hellinger of power spectrum) (mean over 3 seeds, lower is better)



Gap (ortho-vanilla)/vanilla on the mean: PSE (Hellinger of power spectrum)

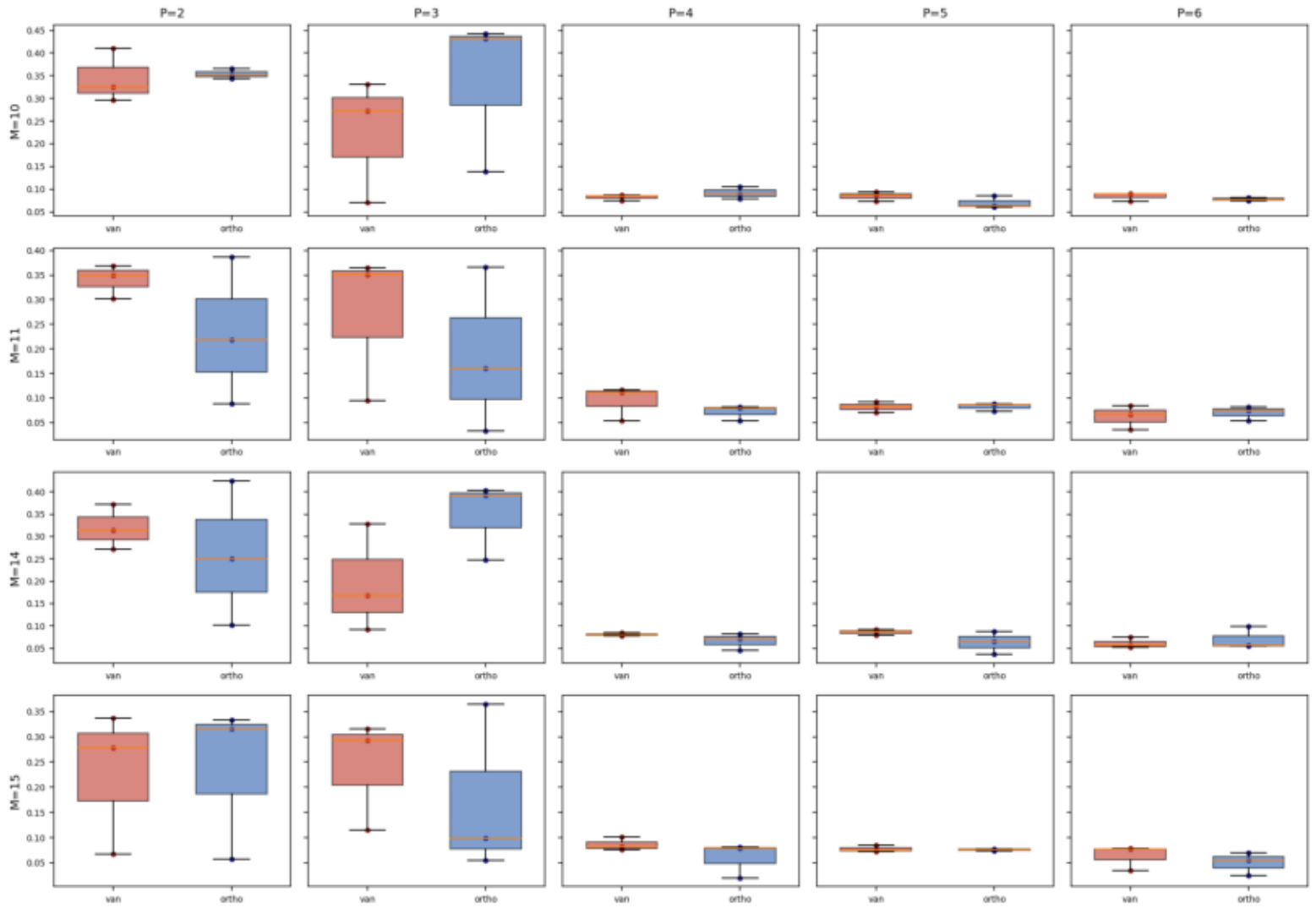


Ortho win rate over 3 seeds: PSE (Hellinger of power spectrum)



Per-cell box plots over 3 seeds: PSE (Hellinger of power spectrum)

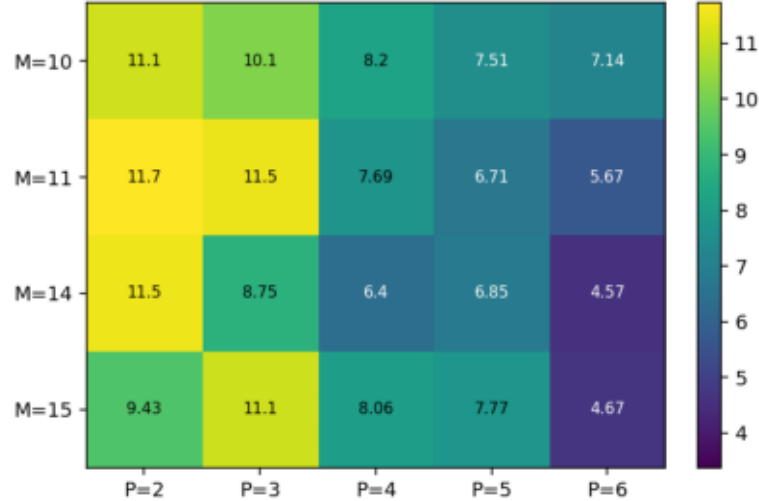
PSE (Hellinger of power spectrum) (per-seed values; box = IQR, line = median)



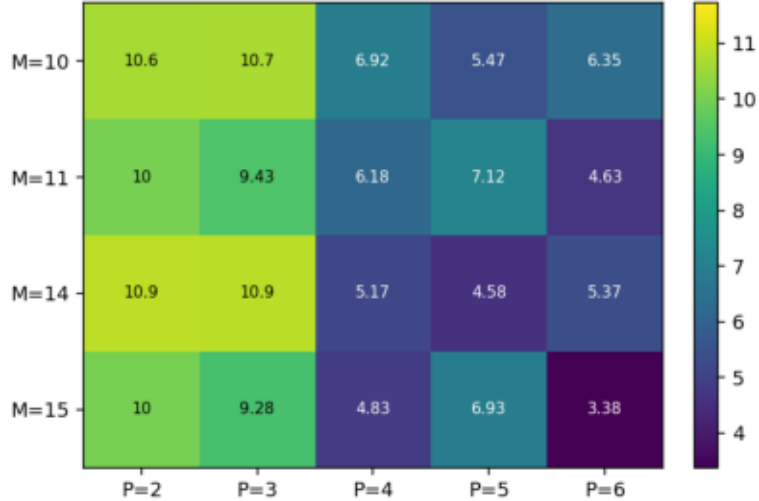
Mean over 3 seeds: Dstsp (KL state-space)

Dstsp (KL) (mean over 3 seeds, lower is better)

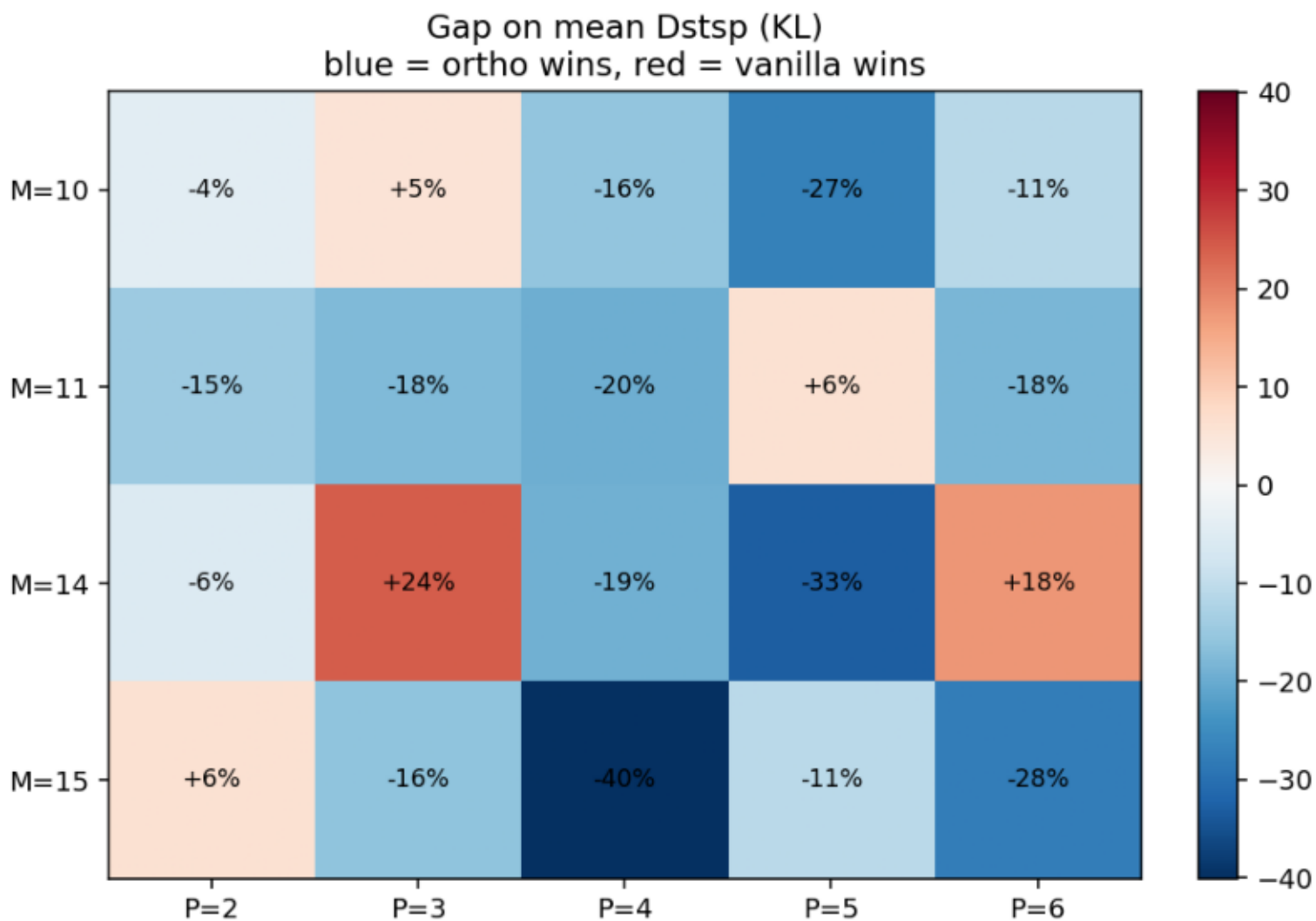
Vanilla mean: Dstsp (KL)



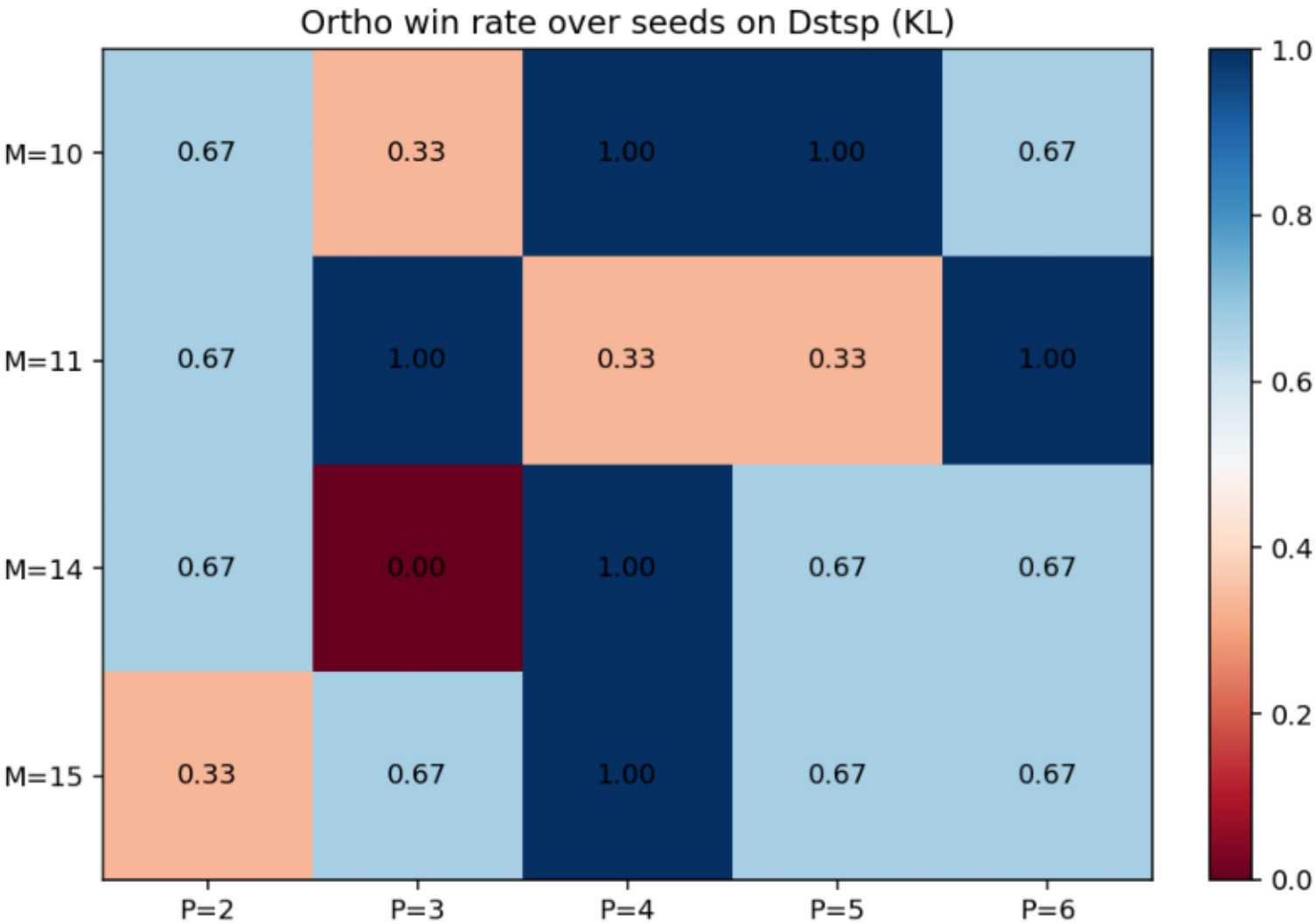
Orthogonal mean: Dstsp (KL)



Gap (ortho-vanilla)/vanilla on the mean: Dstsp (KL state-space)

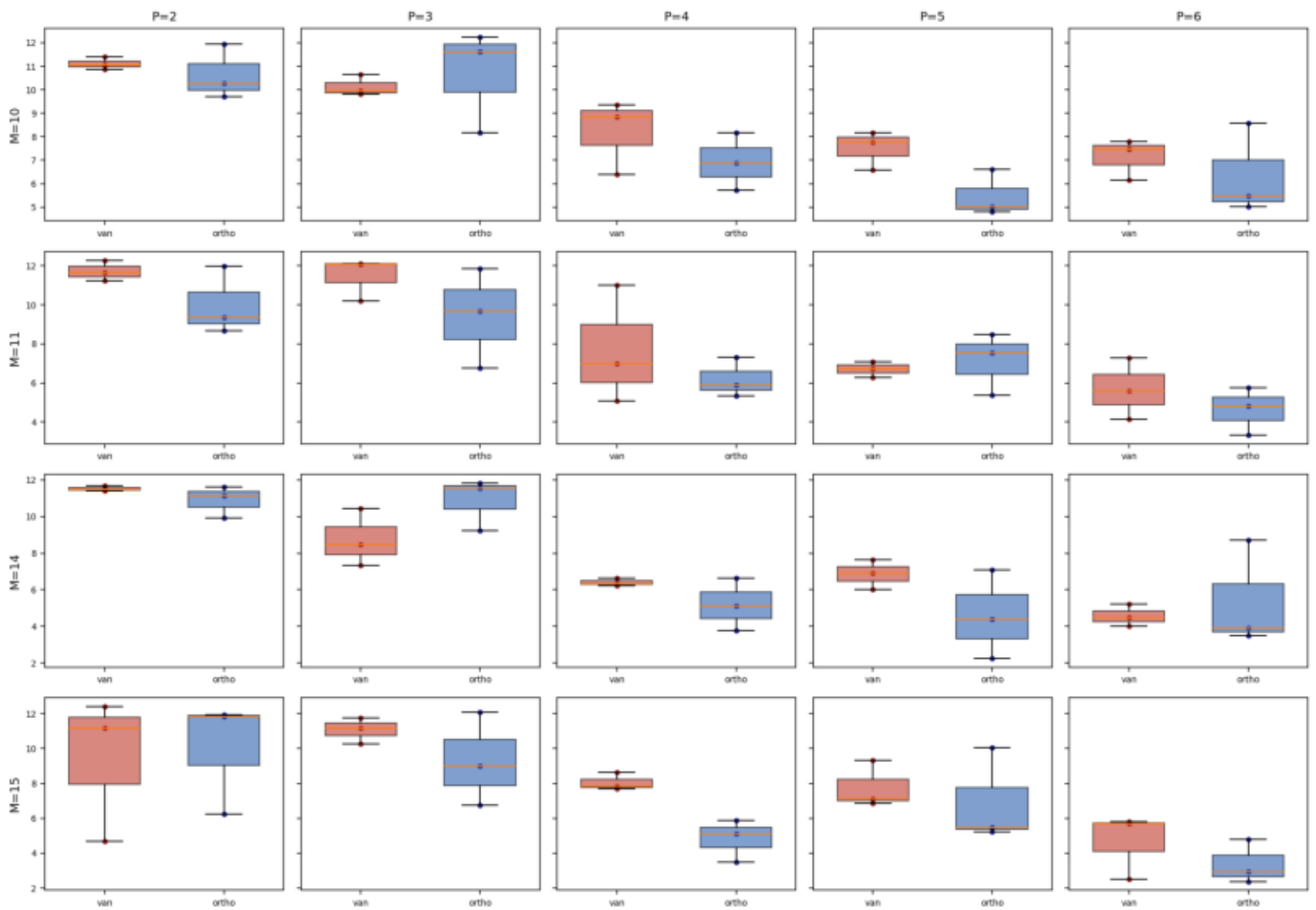


Ortho win rate over 3 seeds: Dstsp (KL state-space)



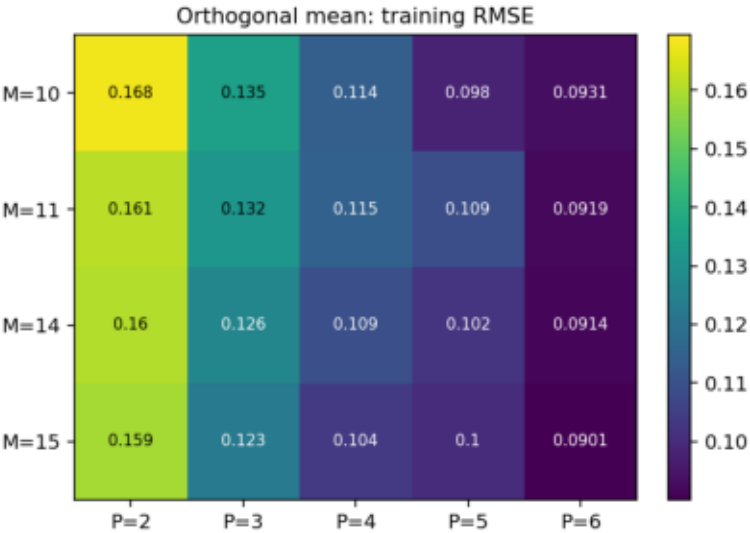
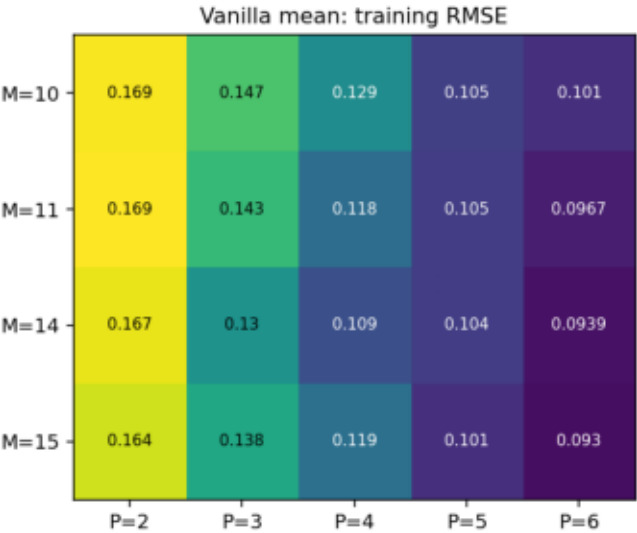
Per-cell box plots over 3 seeds: Dstsp (KL state-space)

Dstsp (KL) (per-seed values; box = IQR, line = median)

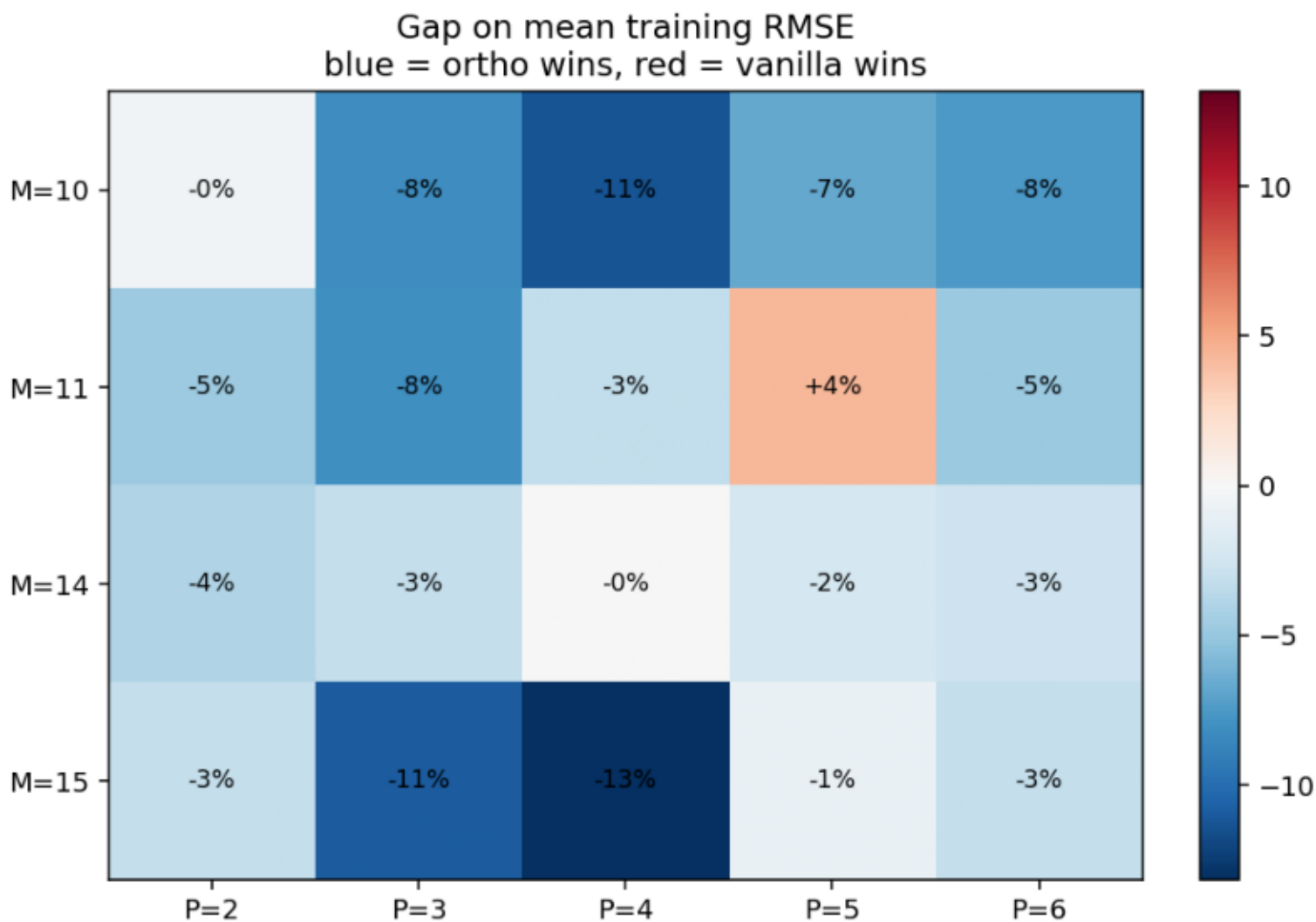


Mean over 3 seeds: training RMSE (one-step)

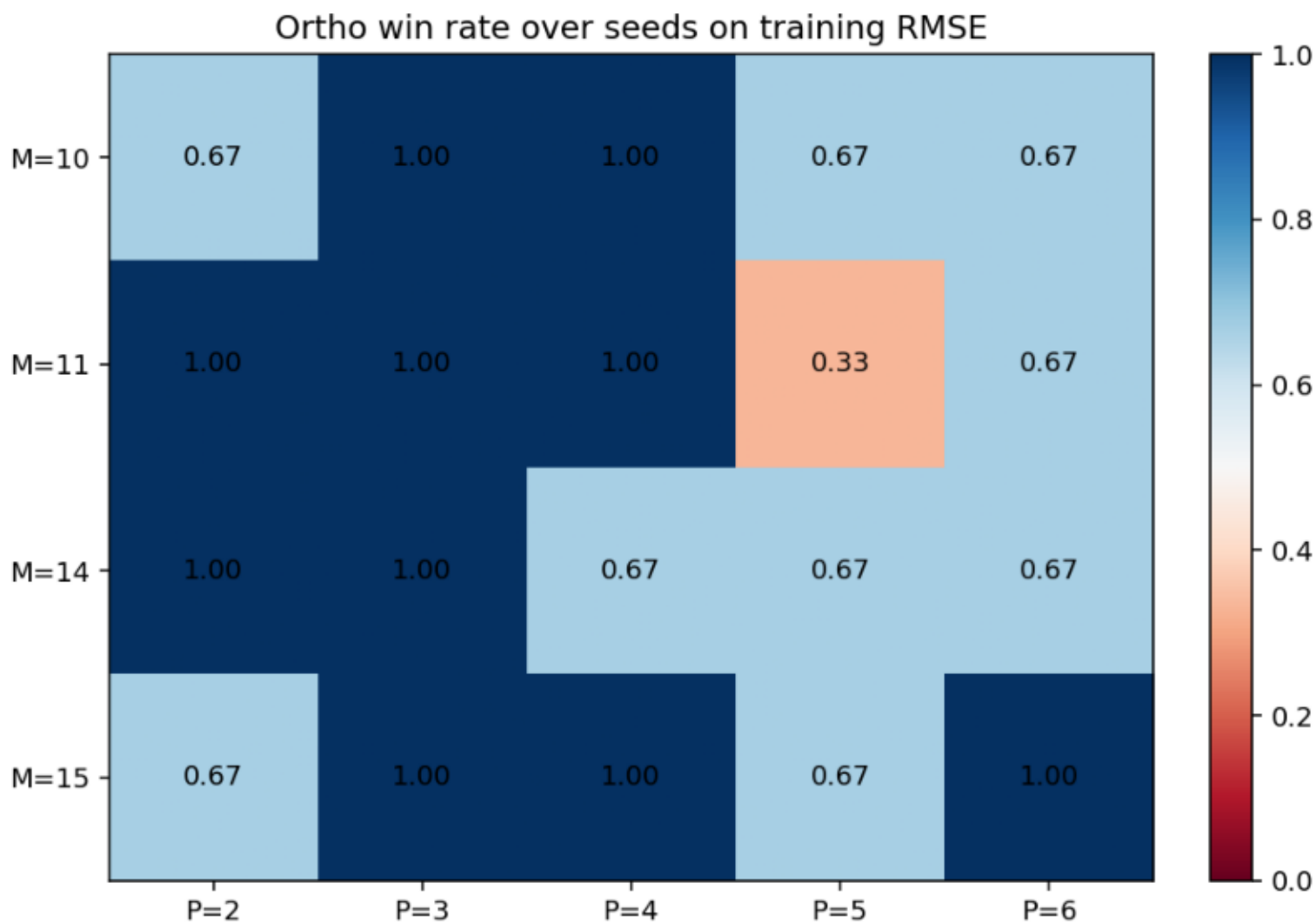
training RMSE (mean over 3 seeds, lower is better)



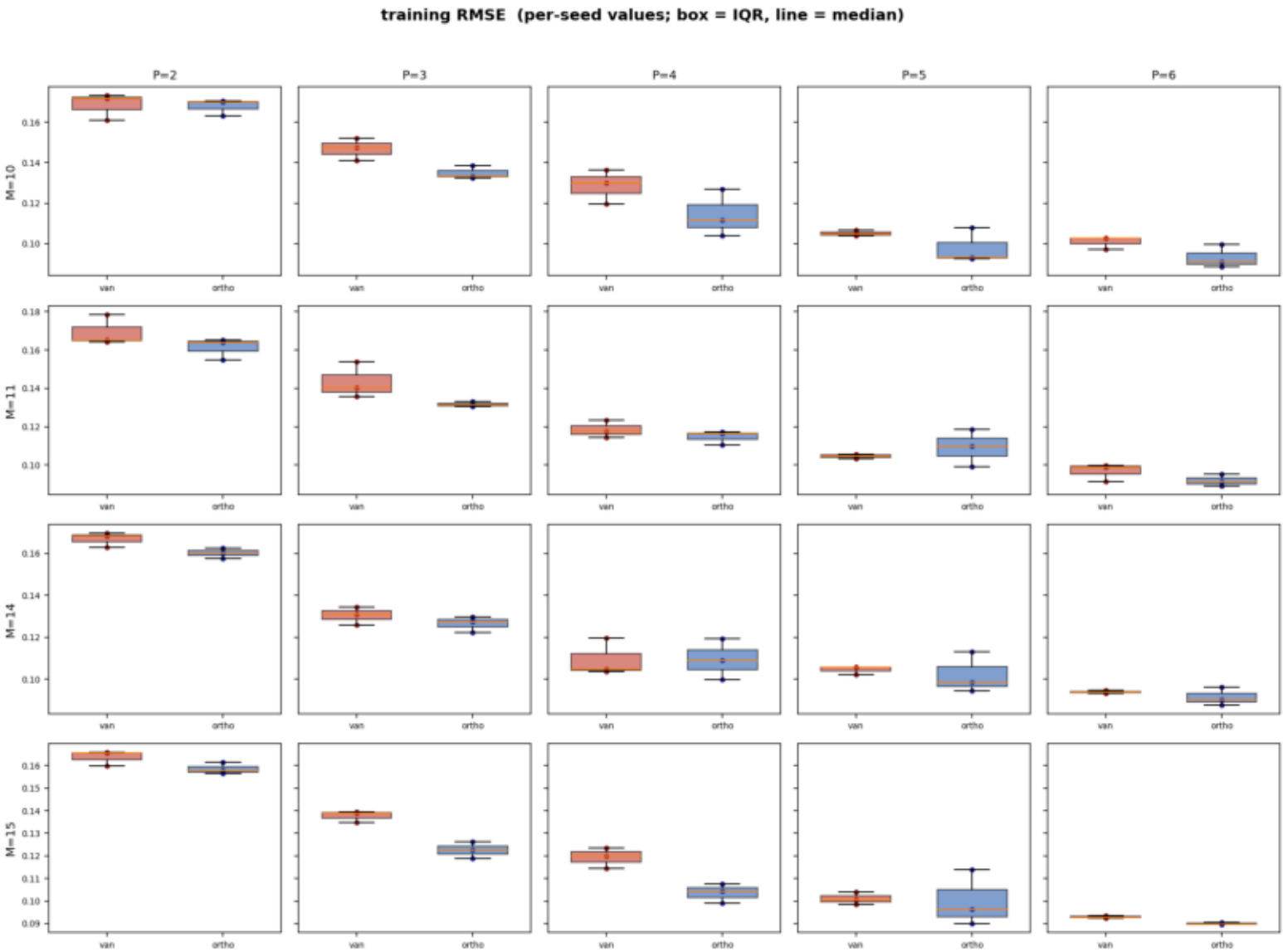
Gap (ortho-vanilla)/vanilla on the mean: training RMSE (one-step)



Ortho win rate over 3 seeds: training RMSE (one-step)

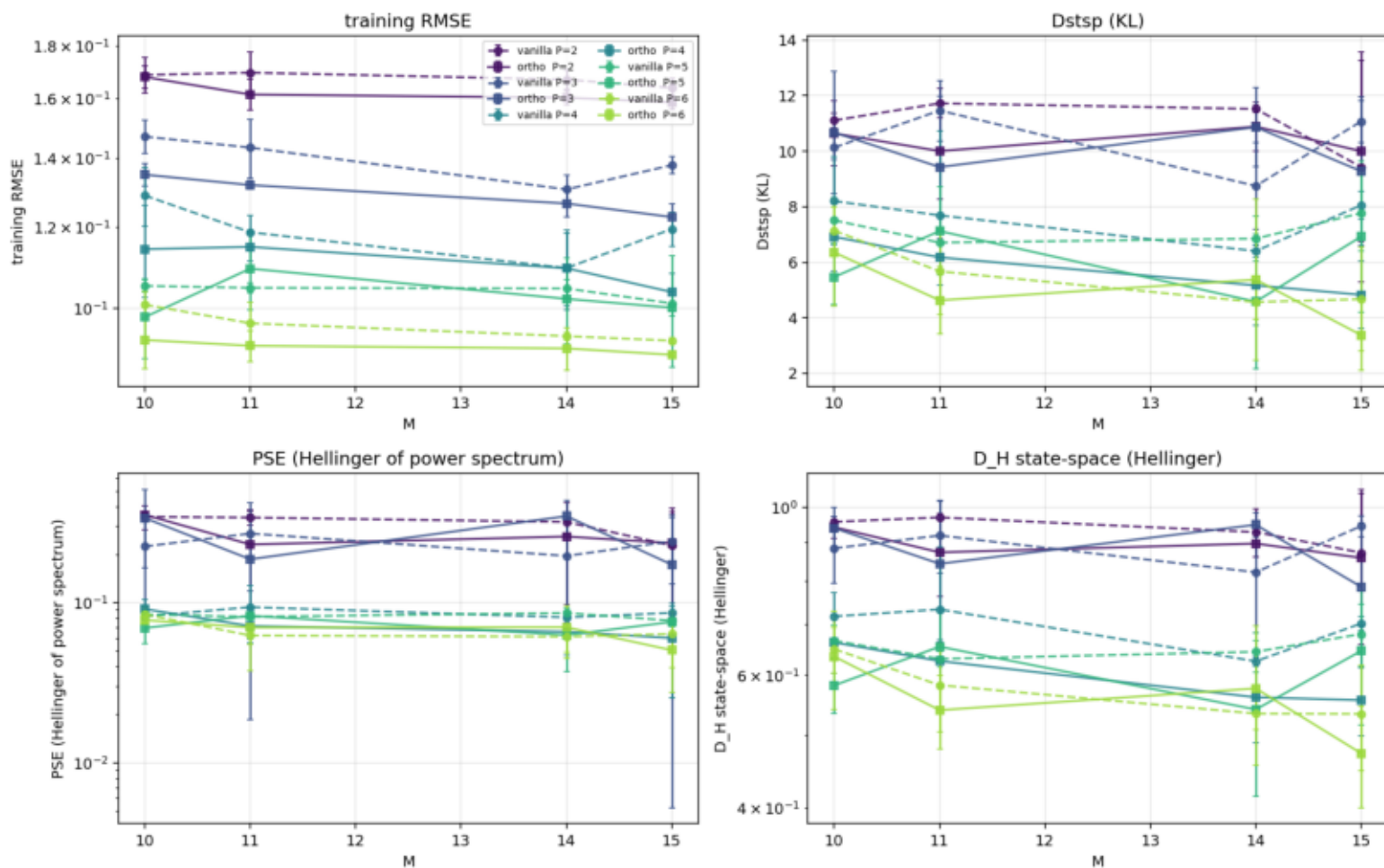


Per-cell box plots over 3 seeds: training RMSE (one-step)



Mean +/- std vs M, by P (vanilla dashed, ortho solid)

Metrics vs M, mean +/- std over 3 seeds (vanilla dashed, ortho solid)



Statistics: D_H_state (priority Hellinger metric)

Per-cell paired statistics on D_H_state (lower is better)

M	P	vanilla	ortho	rel_gap%	d_z	95% t-CI on diff	wilc p	wins
10	2	0.9575±0.007	0.9420±0.031	-1.6%	-0.42	[-0.1076, +0.0766]	0.375	2/3
10	3	0.8833±0.089	0.9394±0.062	+6.4%	+0.40	[-0.2887, +0.4009]	0.750	1/3
10	4	0.7177±0.054	0.6627±0.060	-7.7%	-1.13	[-0.1755, +0.06556]	0.125	3/3
10	5	0.6662±0.033	0.5814±0.048	-12.7%	-3.05	[-0.1538, -0.01578]	0.125	3/3
10	6	0.6493±0.045	0.6349±0.095	-2.2%	-0.17	[-0.223, +0.1942]	0.375	2/3
11	2	0.9706±0.010	0.8725±0.110	-10.1%	-0.84	[-0.3888, +0.1926]	0.250	2/3
11	3	0.9185±0.101	0.8425±0.180	-8.3%	-0.92	[-0.28, +0.1281]	0.125	3/3
11	4	0.7331±0.155	0.6262±0.042	-14.6%	-0.84	[-0.424, +0.2102]	0.250	2/3
11	5	0.6303±0.015	0.6543±0.072	+3.8%	+0.32	[-0.1597, +0.2076]	0.875	1/3
11	6	0.5818±0.078	0.5390±0.060	-7.4%	-1.79	[-0.1023, +0.01672]	0.125	3/3
14	2	0.9281±0.068	0.8961±0.035	-3.4%	-0.39	[-0.2355, +0.1716]	0.375	2/3
14	3	0.8211±0.139	0.9491±0.035	+15.6%	+1.21	[-0.1347, +0.3906]	1.000	0/3
14	4	0.6258±0.009	0.5607±0.072	-10.4%	-0.99	[-0.2278, +0.09763]	0.125	3/3
14	5	0.6441±0.039	0.5401±0.126	-16.1%	-0.66	[-0.4942, +0.2863]	0.250	2/3
14	6	0.5334±0.026	0.5760±0.121	+8.0%	+0.30	[-0.3148, +0.4]	0.625	2/3
15	2	0.8715±0.173	0.8579±0.199	-1.6%	-0.47	[-0.08574, +0.05844]	0.375	2/3
15	3	0.9453±0.030	0.7853±0.170	-16.9%	-0.92	[-0.5899, +0.2698]	0.250	2/3
15	4	0.7022±0.014	0.5557±0.058	-20.9%	-2.55	[-0.2892, -0.003723]	0.125	3/3
15	5	0.6800±0.064	0.6458±0.130	-5.0%	-0.19	[-0.4722, +0.4039]	0.375	2/3
15	6	0.5329±0.085	0.4725±0.072	-11.3%	-0.59	[-0.3138, +0.1931]	0.250	2/3

Pooled across 20 cells

```

mean rel gap           : -5.83%
bootstrap 95% CI on mean rel gap: [-9.66%, -1.78%]
ortho wins on mean      : 16/20
sign-test p (one-sided H1) : 0.0059
Wilcoxon on cell gaps p (H1) : 0.0060

```

Stratified mean rel gaps

```

low_M_only_M10_M11    = -5.44% (8/10 cells)
high_M_only_M14_M15   = -6.21% (8/10 cells)
P_eq_2                 = -4.19% (4/4 cells)
P_eq_3                 = -0.81% (2/4 cells)
P_eq_4                 = -13.38% (4/4 cells)
P_eq_5                 = -7.52% (3/4 cells)
P_eq_6                 = -3.23% (3/4 cells)

```

Statistics: PSE (Hellinger of power spectrum)

Per-cell paired statistics on PSE (lower is better)

M	P	vanilla	ortho	rel_gap%	d_z	95% t-CI on diff	wilc p	wins
10	2	0.3442±0.060	0.3538±0.012	+2.8%	+0.19	[-0.1166, +0.1358]	0.750	1/3
10	3	0.2245±0.137	0.3376±0.172	+50.4%	+0.40	[-0.5954, +0.8217]	0.750	1/3
10	4	0.0822±0.007	0.0912±0.013	+10.9%	+1.12	[-0.01093, +0.02891]	1.000	0/3
10	5	0.0844±0.010	0.0694±0.014	-17.8%	-1.90	[-0.03462, +0.004586]	0.125	3/3
10	6	0.0844±0.009	0.0778±0.004	-7.9%	-1.00	[-0.02312, +0.009864]	0.250	2/3
11	2	0.3403±0.034	0.2310±0.149	-32.1%	-0.62	[-0.5467, +0.3281]	0.250	2/3
11	3	0.2705±0.152	0.1868±0.168	-31.0%	-0.86	[-0.3266, +0.1592]	0.250	2/3
11	4	0.0938±0.034	0.0714±0.015	-23.9%	-0.49	[-0.1354, +0.09065]	0.250	2/3
11	5	0.0816±0.011	0.0828±0.008	+1.4%	+0.06	[-0.04556, +0.0479]	0.625	1/3
11	6	0.0623±0.025	0.0699±0.015	+12.2%	+0.50	[-0.0301, +0.04531]	0.875	1/3
14	2	0.3195±0.051	0.2588±0.162	-19.0%	-0.29	[-0.5877, +0.4664]	0.375	2/3
14	3	0.1959±0.121	0.3473±0.086	+77.3%	+1.75	[-0.06303, +0.3659]	1.000	0/3
14	4	0.0809±0.003	0.0658±0.019	-18.7%	-0.68	[-0.07008, +0.03989]	0.250	2/3
14	5	0.0862±0.006	0.0627±0.026	-27.2%	-0.80	[-0.09653, +0.04958]	0.250	2/3
14	6	0.0613±0.012	0.0702±0.025	+14.5%	+0.26	[-0.07632, +0.0941]	0.750	1/3
15	2	0.2275±0.142	0.2355±0.155	+3.5%	+0.30	[-0.05753, +0.0735]	0.625	2/3
15	3	0.2415±0.110	0.1728±0.168	-28.4%	-0.25	[-0.7587, +0.6213]	0.375	2/3
15	4	0.0864±0.013	0.0600±0.035	-30.5%	-0.81	[-0.1068, +0.05407]	0.250	2/3
15	5	0.0772±0.007	0.0757±0.002	-1.9%	-0.29	[-0.014, +0.01111]	0.375	2/3
15	6	0.0637±0.025	0.0504±0.023	-20.8%	-0.37	[-0.1035, +0.07698]	0.375	2/3

Pooled across 20 cells

mean rel gap : -4.30%
 bootstrap 95% CI on mean rel gap: [-15.23%, +8.80%]
 ortho wins on mean : 12/20
 sign-test p (one-sided H1) : 0.2517
 Wilcoxon on cell gaps p (H1) : 0.1012

Stratified mean rel gaps

low_M_only_M10_M11 = -3.48% (5/10 cells)
 high_M_only_M14_M15 = -5.12% (7/10 cells)
 P_eq_2 = -11.20% (2/4 cells)
 P_eq_3 = +17.07% (2/4 cells)
 P_eq_4 = -15.53% (3/4 cells)
 P_eq_5 = -11.37% (3/4 cells)
 P_eq_6 = -0.50% (2/4 cells)

Pooled & stratified results

Pooled and stratified results for each metric.

Pooled across 20 cells (mean of cell-level relative gaps):

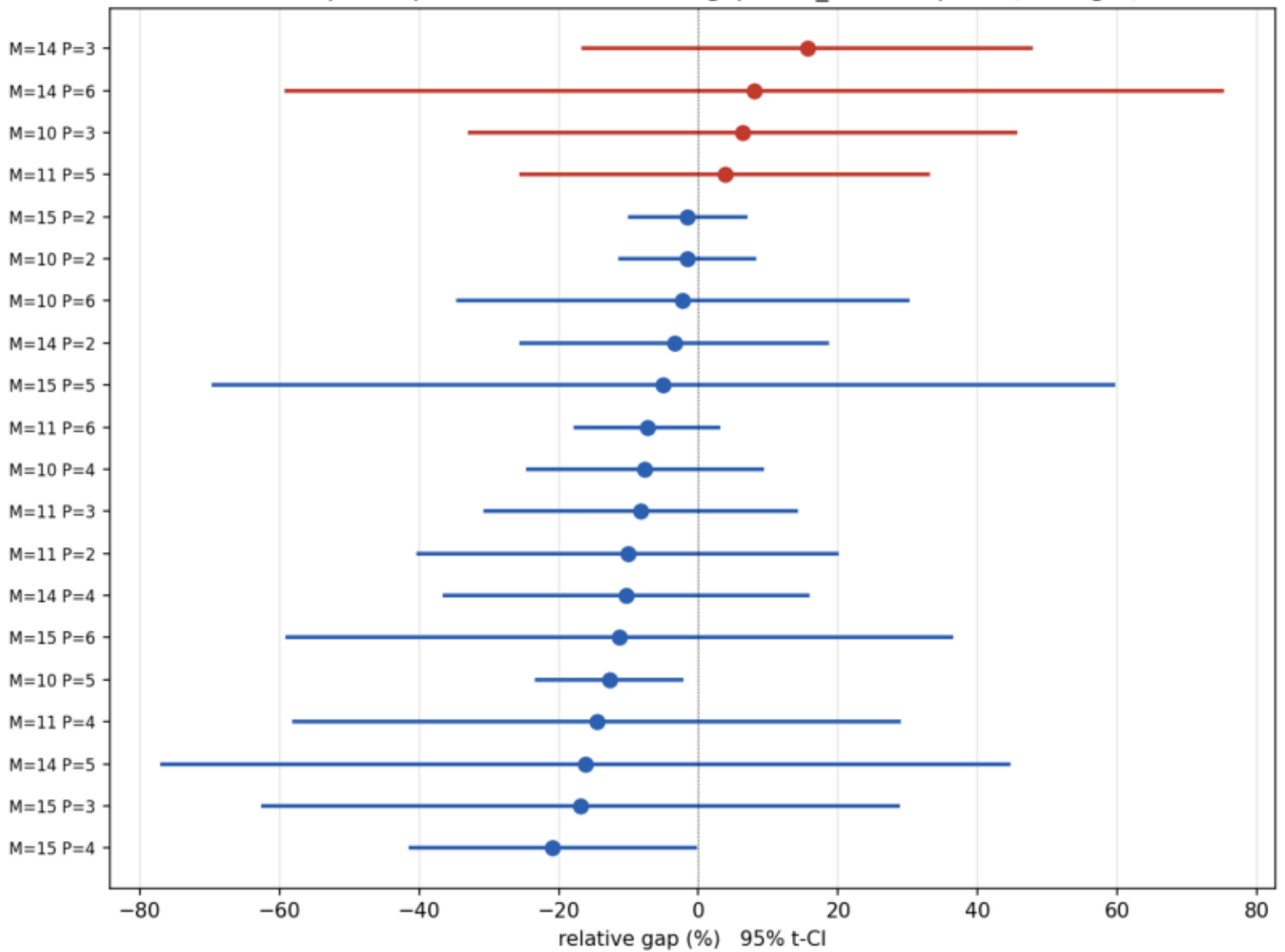
metric	mean rel gap	95% bootstrap CI	wins	sign p	wilc p
D_H_state	-5.83% [-9.66%, -1.78%]	16/20	0.006	0.006	
PSE	-4.30% [-15.23%, +8.80%]	12/20	0.252	0.101	
Dstsp (KL)	-11.09% [-17.99%, -3.91%]	15/20	0.021	0.003	
rmse_train	-4.72% [-6.57%, -2.92%]	19/20	0.000	0.000	

Stratified mean rel gaps (% on mean):

metric	low_M_only_M10_M11	high_M_only_M14_M15	P_eq_2	P_eq_3	P_eq_4
D_H_state	-5.44%	-6.21%	-4.19%	-0.81%	-13.38%
PSE	-3.48%	-5.12%	-11.20%	+17.07%	-15.53%
Dstsp	-11.70%	-10.49%	-4.59%	-1.10%	-23.67%
rmse_train	-5.08%	-4.35%	-3.10%	-7.57%	-6.93%

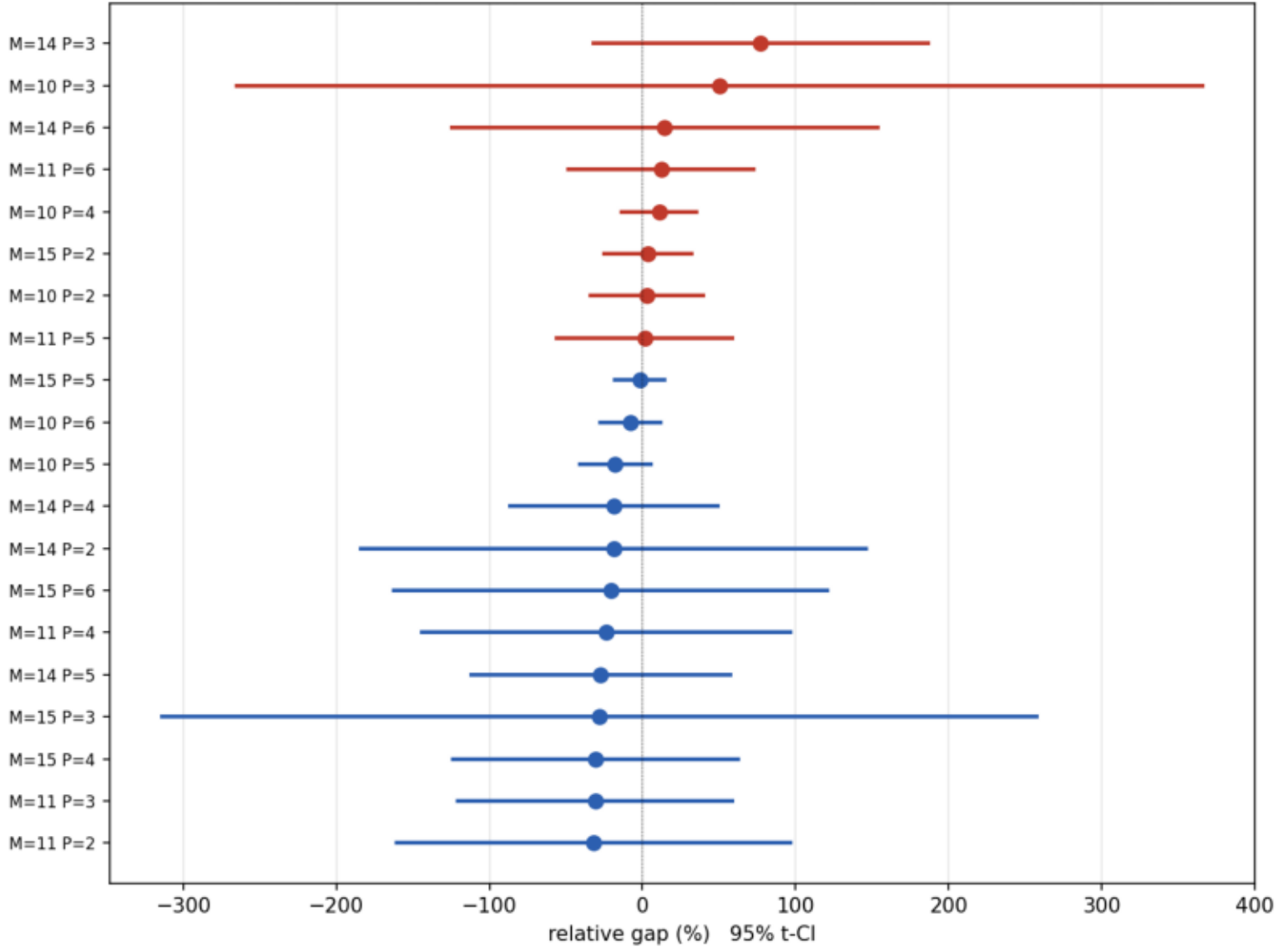
Forest plot: per-cell gap on D_H_state (95% t-CI)

Forest plot of per-cell ortho-vs-vanilla gap on D_H state-space (Hellinger)



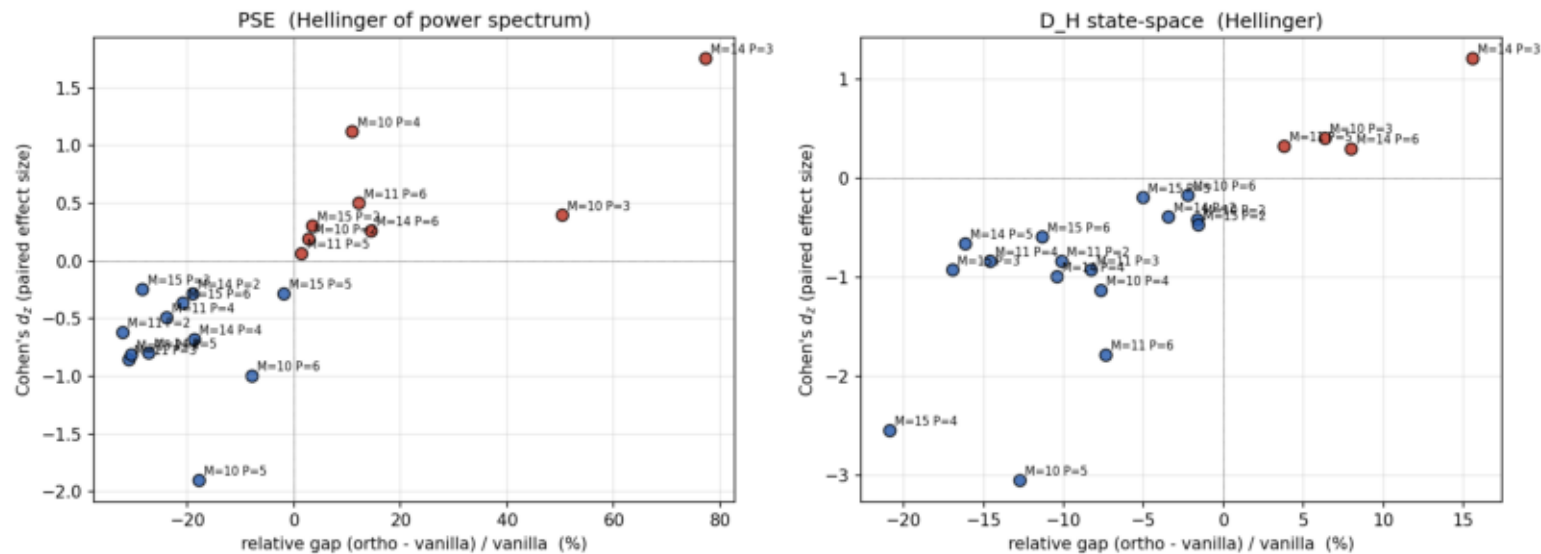
Forest plot: per-cell gap on PSE (95% t-CI)

Forest plot of per-cell ortho-vs-vanilla gap on PSE (Hellinger of power spectrum)



Effect-size scatter: rel gap vs Cohen's d_z (Hellinger metrics)

Per-cell effect sizes on Hellinger metrics (blue = ortho wins on mean)



Highlight trajectories

Each of the next pages shows one (M, P) cell at high resolution: 3 columns of 3-D views (one per seed) plus 3 columns of x-z 2-D projections (one per seed). Rows are ground truth, vanilla AL-RNN, orthogonal AL-RNN.

Ordered by D_H_state gain (largest ortho wins first):

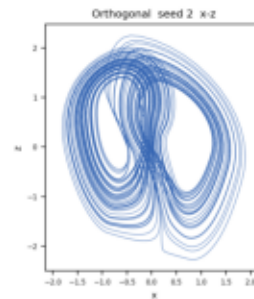
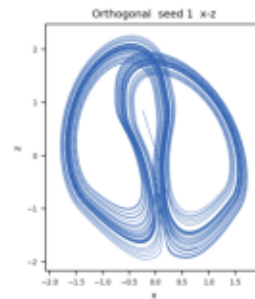
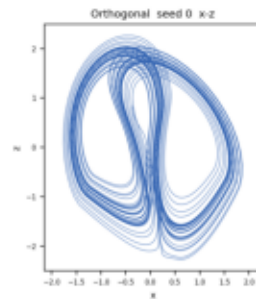
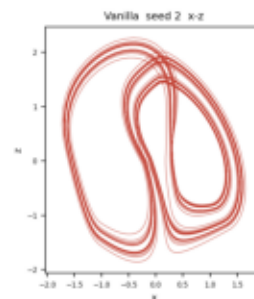
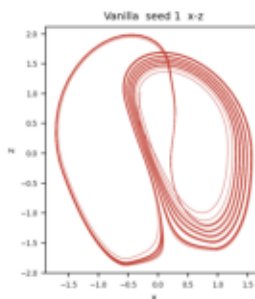
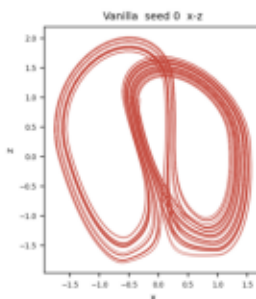
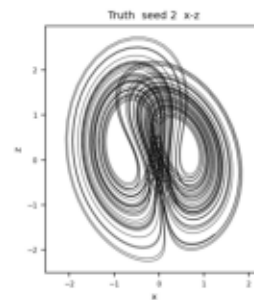
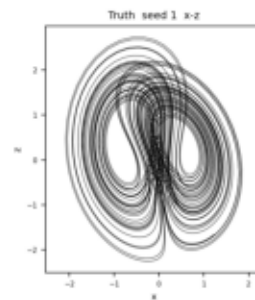
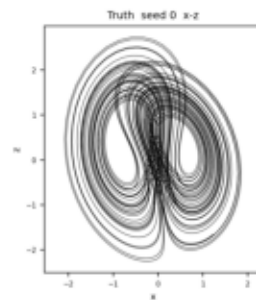
M=15	P=4	D_H_state gap =	-20.9%	PSE gap =	-30.5%
M=15	P=3	D_H_state gap =	-16.9%	PSE gap =	-28.4%
M=14	P=5	D_H_state gap =	-16.1%	PSE gap =	-27.2%
M=11	P=4	D_H_state gap =	-14.6%	PSE gap =	-23.9%
M=10	P=5	D_H_state gap =	-12.7%	PSE gap =	-17.8%
M=15	P=6	D_H_state gap =	-11.3%	PSE gap =	-20.8%

Plus one ortho-failure cell:

M=14	P=3	D_H_state gap =	+15.6%	(largest ortho regression on D_H_state)
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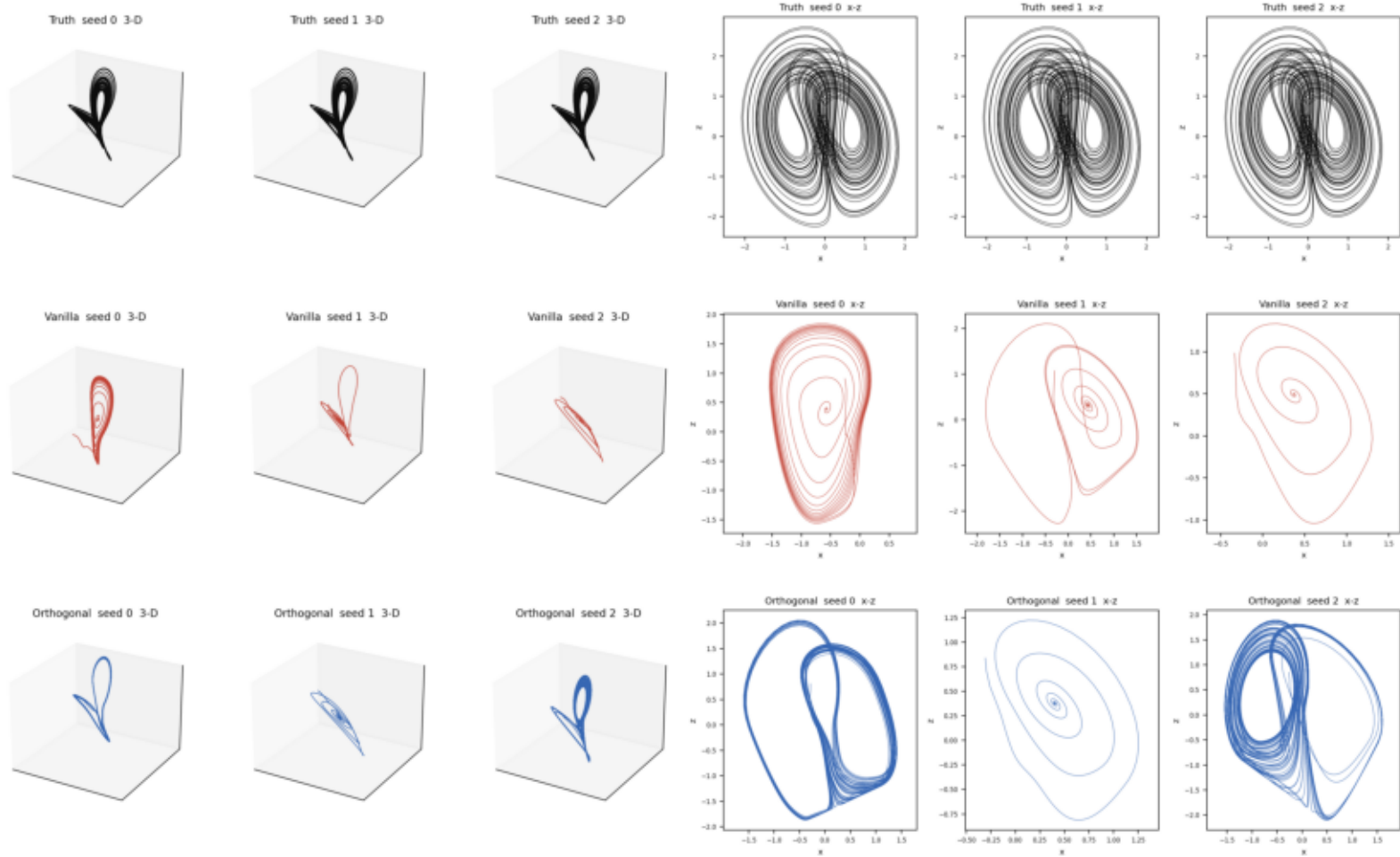
M=15 P=4: ortho wins by -20.9% on D_H_state

PSE vanilla 0.0864 ± 0.0128 ortho 0.0600 ± 0.0345 **M = 15 P = 4** (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D) **D_H_state vanilla 0.702 ± 0.014 ortho 0.556 ± 0.058 mean rel gap -20.9%**



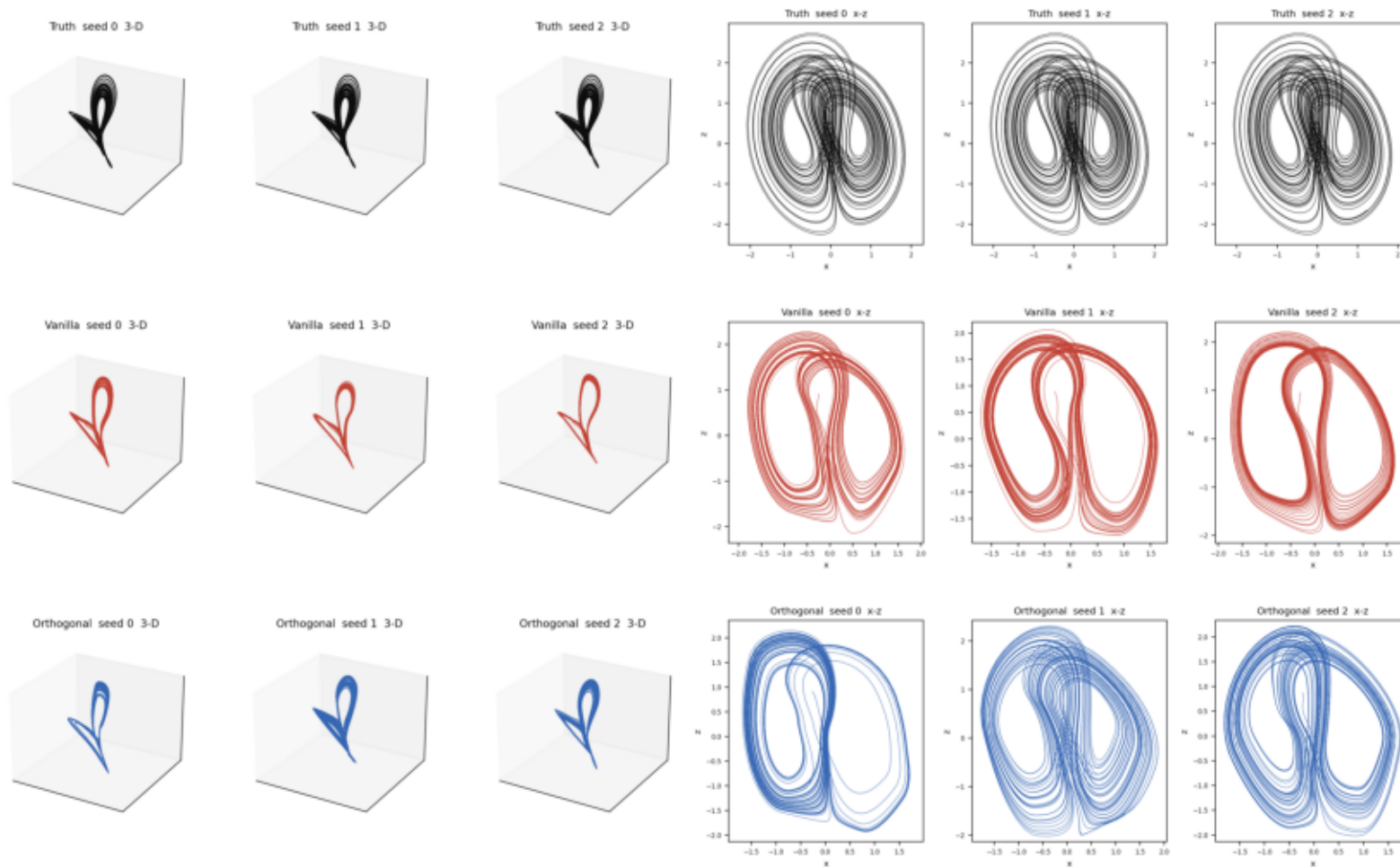
M=15 P=3: ortho wins by -16.9% on D_H_state

PSE vanilla 0.2415 ± 0.1102 ortho 0.1728 ± 0.1676 **M = 15 P = 3** (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D) **D_H_state vanilla 0.945 ± 0.030 ortho 0.785 ± 0.170 mean rel gap -16.9%**



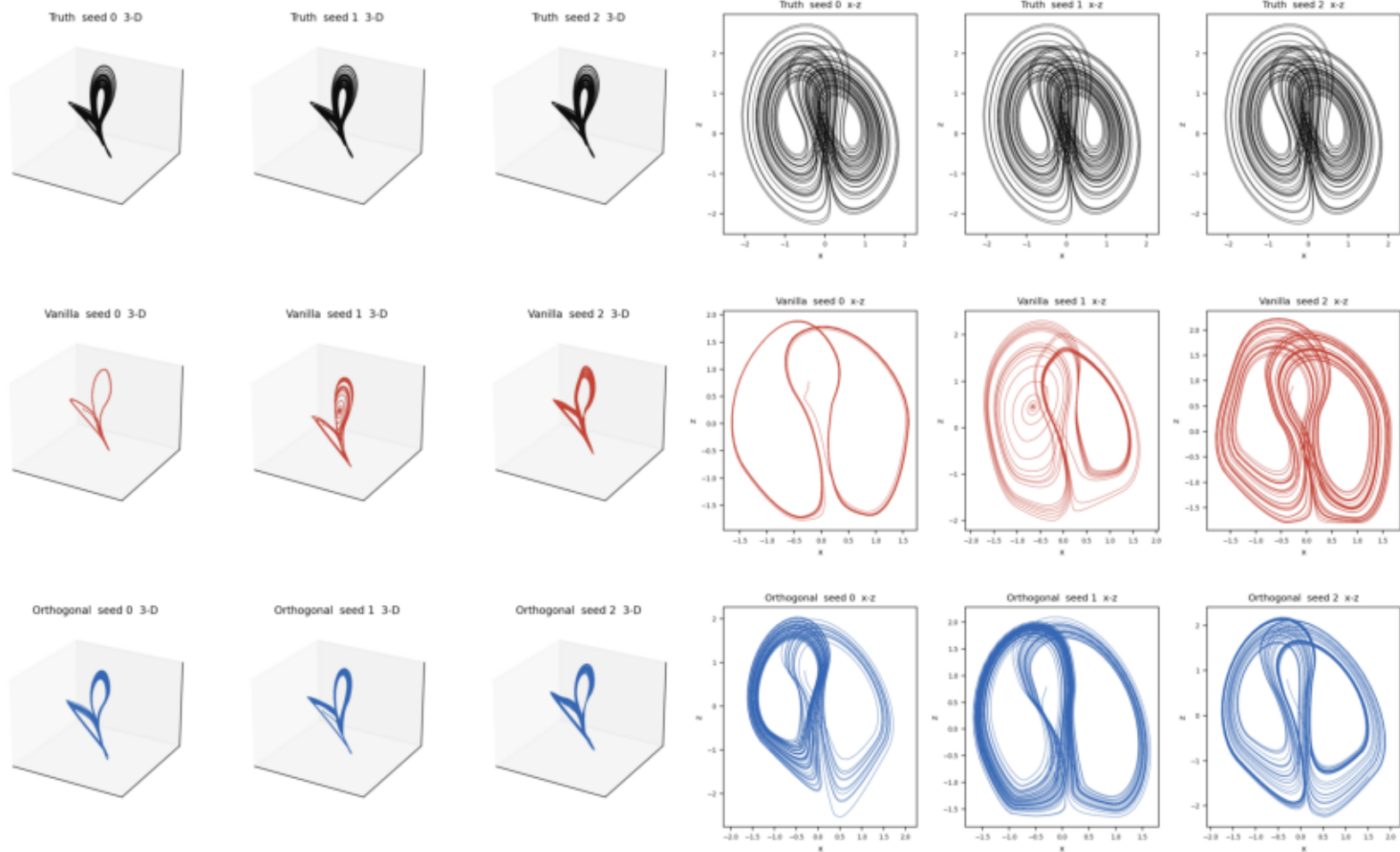
M=14 P=5: ortho wins by -16.1% on D_H_state

PSE vanilla 0.0862 ± 0.0061 ortho 0.0627 ± 0.0256 **M = 14 P = 5** (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D) **D_H_state vanilla 0.644 ± 0.039 ortho 0.540 ± 0.126 mean rel gap -16.1%**



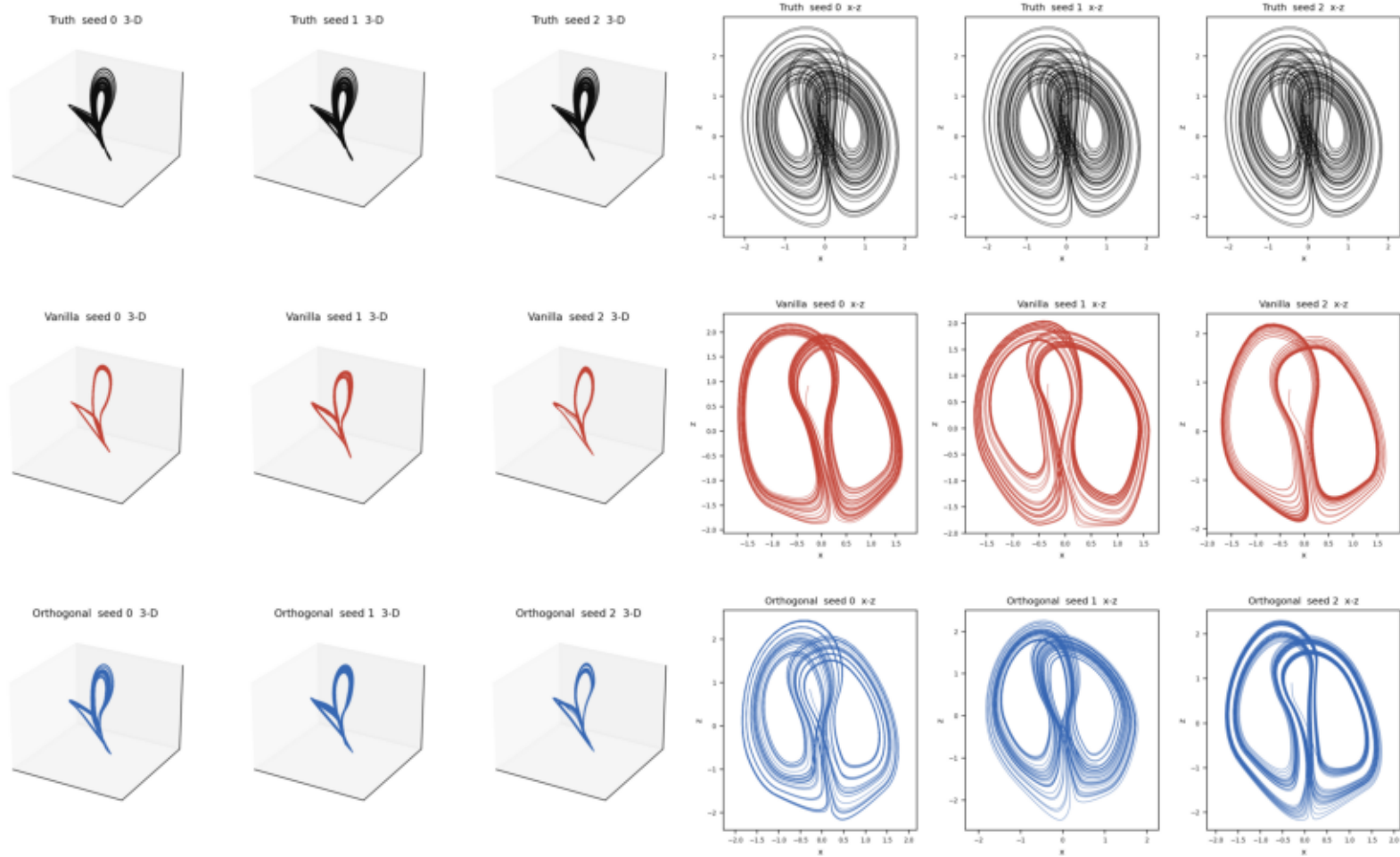
M=11 P=4: ortho wins by -14.6% on D_H_state

PSE vanilla 0.0938 ± 0.0345 ortho 0.0714 ± 0.0151 **M = 11 P = 4 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)** mean rel gap -23.9% **D_H_state vanilla 0.733 ± 0.155 ortho 0.626 ± 0.042 mean rel gap -14.6%**



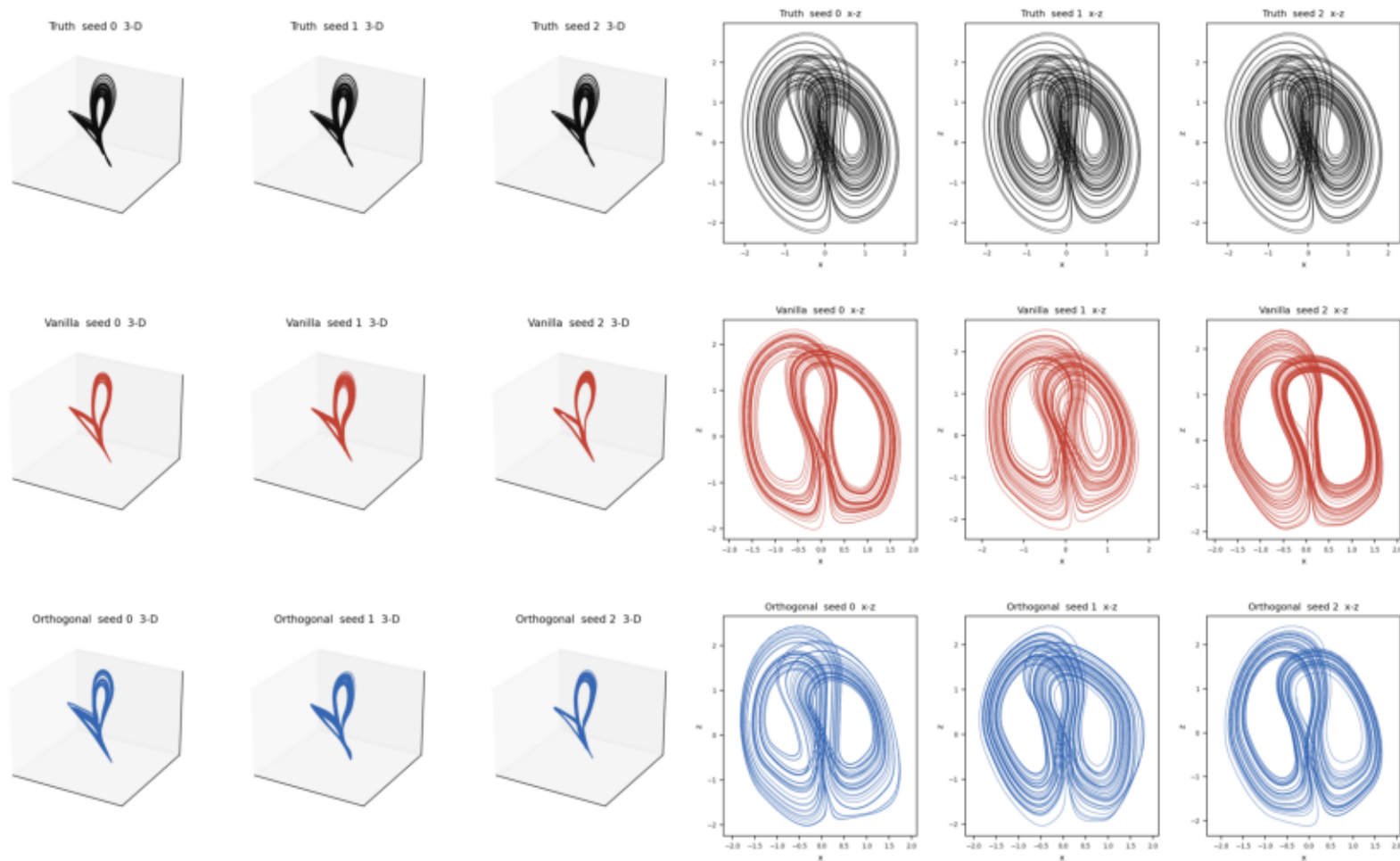
M=10 P=5: ortho wins by -12.7% on D_H_state

PSE vanilla 0.0844 ± 0.0103 ortho 0.0694 ± 0.0139 M = 10 P = 5 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D) mean rel gap -17.8% D_H_state vanilla 0.666 ± 0.033 ortho 0.581 ± 0.048 mean rel gap -12.7%



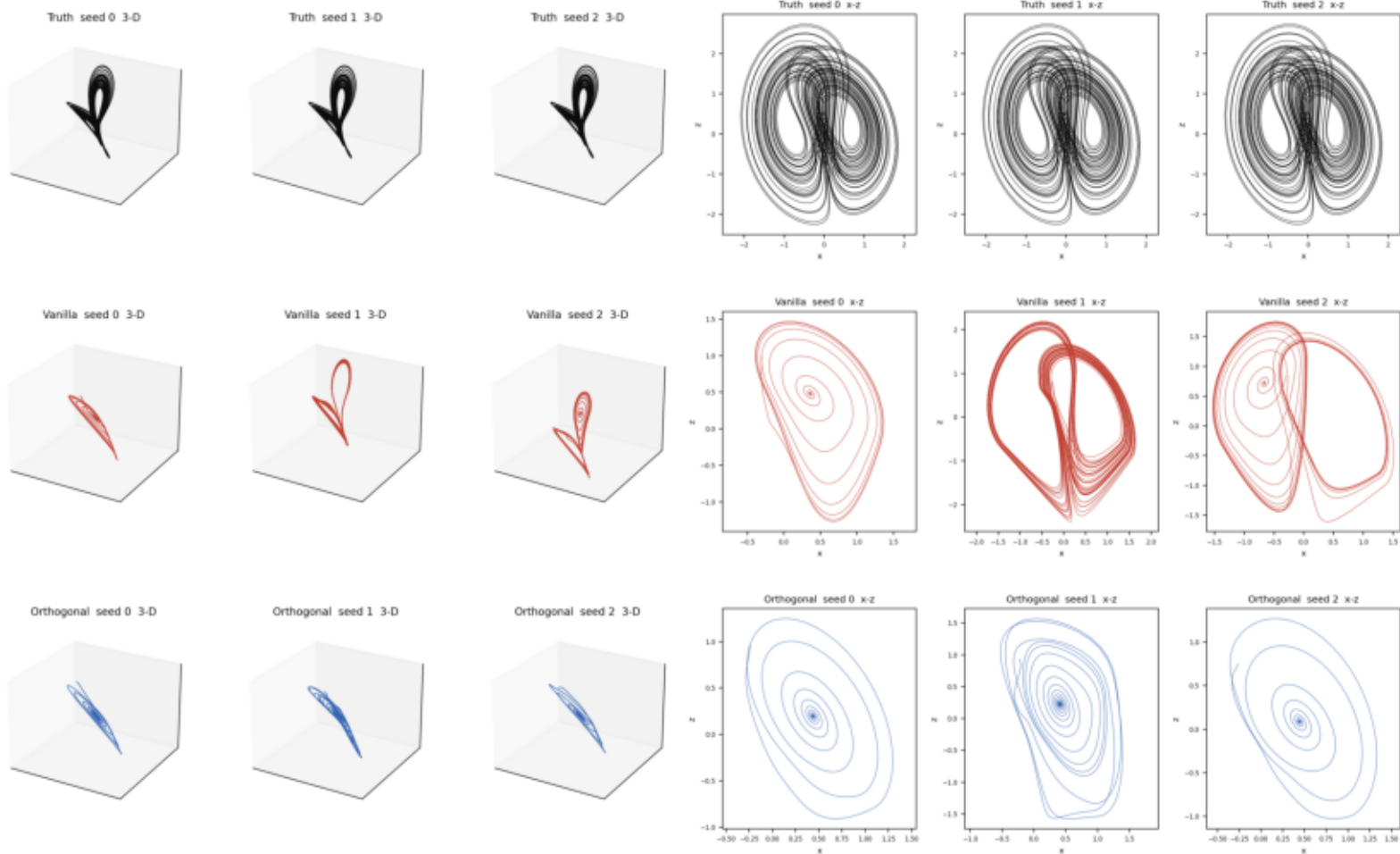
M=15 P=6: ortho wins by -11.3% on D_H_state

PSE vanilla 0.0637 ± 0.0247 ortho 0.0504 ± 0.0230 M = 15 P = 6 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D) mean rel gap -20.8% D_H_state vanilla 0.533 ± 0.085 ortho 0.473 ± 0.072 mean rel gap -11.3%



M=14 P=3: ortho regression (+15.6% on D_H_state)

M = 14 P = 3 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE vanilla 0.1959 ± 0.1212 ortho 0.3473 ± 0.0862 mean rel gap +77.3% D_H_state vanilla 0.821 ± 0.139 ortho 0.949 ± 0.035 mean rel gap +15.6%



Trajectory wall: all 20 cells, median-PSE seed, x-z projection

Trajectory wall: all (M, P) cells, median-PSE seed, x-z projections

